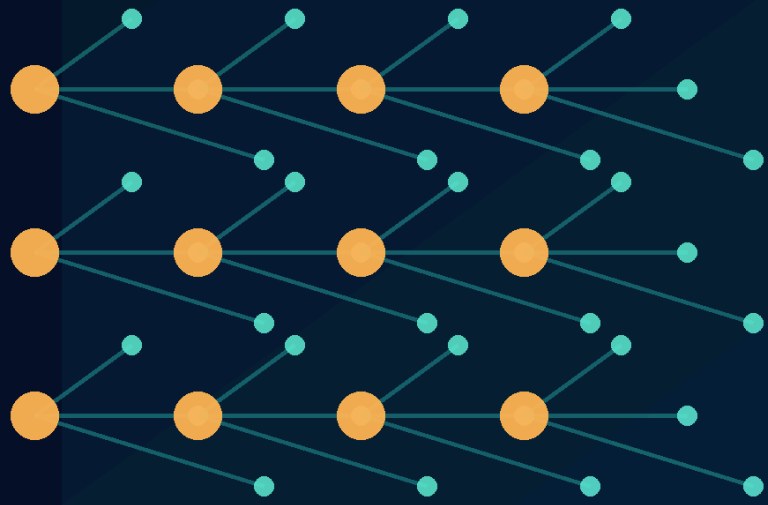
BUKU TUTORIAL | EDISI 2.0

OTP-AI

MACHINE LEARNING DENGAN PASCAL



Panduan tutorial 20 program contoh untuk Odin Turbo Pascal

Penulis

Syams Ideris

GNU GPL versi 3.0 atau yang lebih baru

19 UNIT • 20 CONTOH • LANGKAH PRAKTIS

Hak cipta dan lisensi

OTP-AI — Machine Learning dengan Pascal

Penulis: Syams Ideris

Hak cipta (c) 2026 Syams Ideris.

Buku ini dilisensikan berdasarkan GNU General Public License versi 3.0, atau (sesuai pilihan Anda) versi yang lebih baru (GPL-3.0-or-later). Teks lisensi lengkap disertakan pada bagian akhir buku dan di file LICENCE.TXT.

> **Pemberitahuan merek dan kompatibilitas**
>
> Odin Turbo Pascal adalah nama IDE/compiler pada proyek ini. Turbo Pascal 7.0 dan Free Pascal dibahas sebagai target/compiler terpisah; hasil pengujian satu compiler tidak otomatis membuktikan kompatibilitas compiler lain.

Buku menggunakan source contoh dalam paket otp-AI. Gambar editor pada tutorial merupakan visualisasi layar yang dibuat dari source aktual; tata letak menu dapat berbeda menurut build IDE. Angka hasil hanya dicantumkan jika bersumber dari contoh atau laporan pengujian paket.

Edisi: OTP-AI 2.0.0. Untuk perubahan API, baca DOCS/API.TXT dan DOCS/DESIGN.TXT di dalam paket.

PEMBUKA

Kata pengantar

Machine learning sering terasa jauh karena biasanya diperkenalkan lewat library besar, notebook, atau layanan daring. Buku ini memilih jalan yang berbeda: setiap langkah penting diperlihatkan dengan Pascal, data kecil, dan program yang dapat dibaca dari awal hingga akhir.

OTP-AI berisi 19 unit Pascal untuk matematika dan matriks, data, preprocessing, validasi, regresi, klasifikasi, neural network kecil, teks, pohon, aturan, pencarian, fuzzy, reinforcement learning, plotting, serta penyimpanan model. Dua puluh program contoh membawa pembaca dari operasi numerik dasar menuju model AI yang lebih kaya.

Tujuan buku bukan menjanjikan model siap produksi. Batas ukuran, asumsi algoritma, status kompilasi, dan hal yang belum diuji ditulis secara terbuka. Data contoh bersifat sintetis; ia baik untuk belajar dan buruk sebagai bukti mutu pada data nyata.

Ikuti latihan dengan mengubah satu parameter setiap kali. Baca kode sebelum menekan Run, catat prediksi Anda, lalu bandingkan hasil. Jika satu contoh gagal, periksa pesan error dan working directory sebelum mengubah model.

Selamat belajar dan bereksperimen.

Syams Ideris

NAVIGASI

Daftar isi

Bab tutorial memakai source terkini dari folder EXAMPLES.

01 BASIC — Matematika, matriks, dan statistik 7

02 DATAPREP — Membaca, membagi, dan mempersiapkan data 9

03 REGDEMO — Regresi linear satu fitur 11

04 REGMULT — Regresi multivariat dan koefisien 13

05 KNNDEMO — K-nearest neighbors untuk klasifikasi 15

06 KMEANS — Clustering tanpa label 17

07 CROSSVAL — Validasi silang K-fold 19

08 LOGDEMO — Regresi logistik 21

09 PERDEMO — Perceptron dan batas linear 23

10 NNDEMO — Jaringan saraf kecil untuk XOR 25

Peta unit dan referensi

Lampiran A — Peta 19 unit 47

Lampiran B — Diagnostik dan batas 48

GNU GPL versi 3.0 atau yang lebih baru 49+

NAVIGASI

Daftar isi · lanjutan

11 BAYDEMO — Naive Bayes Gaussian 27

12 TEXTDEMO — Teks, token, dan TF-IDF 29

13 TREEDEMO — Pohon keputusan 31

14 RULEDEMO — Sistem aturan dan forward chaining 33

15 PATHDEMO — Pencarian jalur pada grid 35

16 FUZDEMO — Inferensi fuzzy Mamdani 37

17 QDEMO — Q-learning pada dunia grid 39

18 OPTDEMO — Optimizer dan satu langkah SGD 41

19 MODELIO — Menyimpan dan memuat model 43

20 PLOTDEMO — Grafik BGI dan SVG 45

Peta belajar

Ikuti urutan dari BASIC hingga PLOTDEMO. Jika fokus Anda satu topik, gunakan unit canonical di Lampiran A dan API.TXT untuk mencari signature resmi.

Cara menggunakan buku

Siapkan compiler Odin Turbo Pascal dari proyek ini. Buka satu program .PAS dari otp-AI/EXAMPLES, lalu tekan Ctrl+F9. Sembilan belas unit OTP-AI sudah terpasang di IDE, jadi klausa uses AIMATH, AIDATA, AILINEAR; langsung dikenali seperti uses Crt; tanpa mengisi Options > Directories. Contoh membaca berkas DATA dari folder induk secara otomatis, dan berkas hasil ditulis di folder program.

Kompilasi dari terminal:

Perintah Odin

```
mkdir -p BIN && odinpc build EXAMPLES/BASIC.PAS --out BIN/BASIC --unit-path UNITS --include-path EXAMPLES --rtl /path/ke/Odin_Turbo_Pascal/rtl
```

Atau gunakan Free Pascal mode TP untuk memeriksa kompatibilitas source: sh BUILDFPC.SH. Build script mengompilasi contoh dan test, tetapi pada host ini eksekusi/link Free Pascal belum terverifikasi.

Struktur paket

- UNITS — 19 unit Pascal canonical.
- EXAMPLES — 20 tutorial program dan COMMON.INC.
- DATA — fixture CSV/TSV sintetis.
- TESTS — suite Pascal, oracle, dan pembanding numerik opsional.
- DOCS — API, rancangan, panduan, instalasi, serta laporan pengujian.

Urutan aman

Baca klausa uses dan data input, build, jalankan dari root paket, periksa AIError bila fungsi mengembalikan False, lalu ubah hanya satu parameter.

Matematika, matriks, dan statistik

Program BASIC.PAS | Unit: AIMATH + AISTATS

Mulai dari operasi numerik yang menjadi dasar model AI. BASIC menunjukkan validasi input, matriks, dan ringkasan statistik tanpa dataset eksternal.

PERAN UNIT

AIMATH menyediakan operasi numerik, vektor, jarak, dan matriks; AISTATS merangkum data menjadi statistik yang dapat dibaca model.

TUJUAN CONTOH

Menghubungkan angka mentah, operasi aljabar, dan ringkasan statistik sebelum mempelajari estimator.

FUNGSI DAN PARAMETER UTAMA

N=2 untuk Euclidean/DotProduct; matriks A dan identitas B berukuran 2×2; statistik memakai lima nilai dan Variance(Sample=True).

HASIL YANG DIHARAPKAN

Jarak (1,2) ke (4,6)=5; dot product=16; sigmoid(0)=0.5; A×I=A; mean dan median=3; varians sampel=2.5; korelasi=1.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Fondasi ML: jarak dipakai KNN/clustering, dot product membentuk skor model linear, statistik membantu memahami sebaran fitur.

ALUR KODE DAN CATATAN

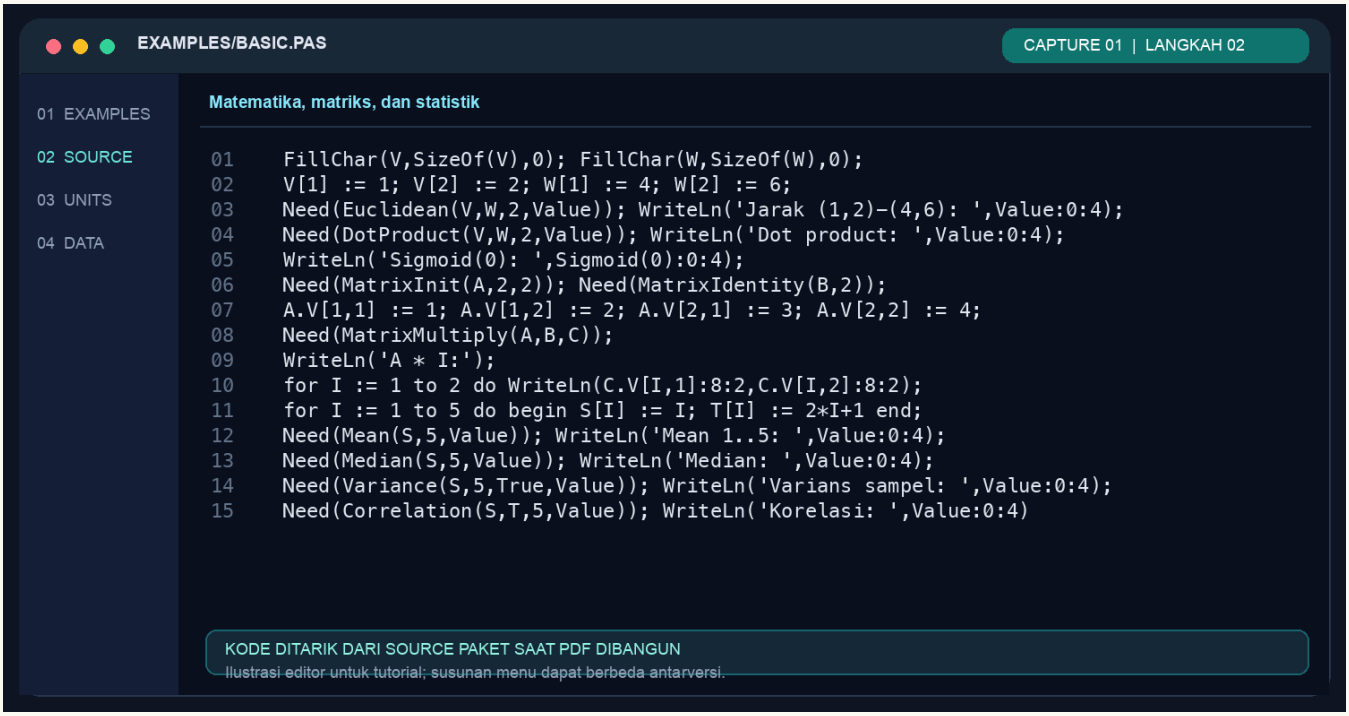
Buat vektor dan matriks kecil; panggil fungsi; periksa Boolean; amati hasil yang dicetak. MatrixMultiply memakai output sementara: pemanggilan in-place aman dan hasil gagal tidak mengganti output.

Rujukan source: EXAMPLES/BASIC.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 01 — BASIC

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Buka EXAMPLES/BASIC.PAS dan periksa klausa uses AIMATH serta AISTATS.
2. Build BASIC.PAS dengan folder UNITS dan EXAMPLES pada search path.
3. Run dari folder otp-AI; bandingkan jarak Euclidean, dot product, sigmoid, dan statistik.
4. Ubah satu nilai input, build ulang, lalu jelaskan perubahan output.



Gambar 01.1 — Visualisasi editor dengan cuplikan source BASIC.PAS.

Periksa hasil: MatrixMultiply memakai output sementara: pemanggilan in-place aman dan hasil gagal tidak mengganti output.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Membaca, membagi, dan mempersiapkan data

Program DATAPREP.PAS | Unit: AIDATA + AISPLIT

Pipeline data memisahkan nilai hilang, imputasi, dan scaling. Parameter preprocessing harus dipelajari dari data latih saja untuk mencegah data leakage.

PERAN UNIT

AIDATA menangani CSV, missing value, imputer, scaler, kategori, dan encoding; AISPLIT membagi baris untuk train dan test.

TUJUAN CONTOH

Membangun preprocessing berurutan pada data yang memiliki nilai hilang, lalu mengekspor hasil agar perubahan bisa diperiksa.

FUNGSI DAN PARAMETER UTAMA

TrainTestSplit fraction=0.25 membagi 25% untuk uji, shuffle=True mengacak, seed=42 mengulang urutan. FitImputer/FitScaler belajar dari train; ScaleStandard memakai mean dan SD.

HASIL YANG DIHARAPKAN

Program mencetak ringkasan dataset, jumlah baris terpilih, encoding MERAH/BIRU, dan menyimpan PREPTRN.CSV serta PREPTST.CSV.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Mencegah data leakage: parameter imputasi dan scaling dihitung dari train saja, kemudian dipakai konsisten untuk test/produksi.

ALUR KODE DAN CATATAN

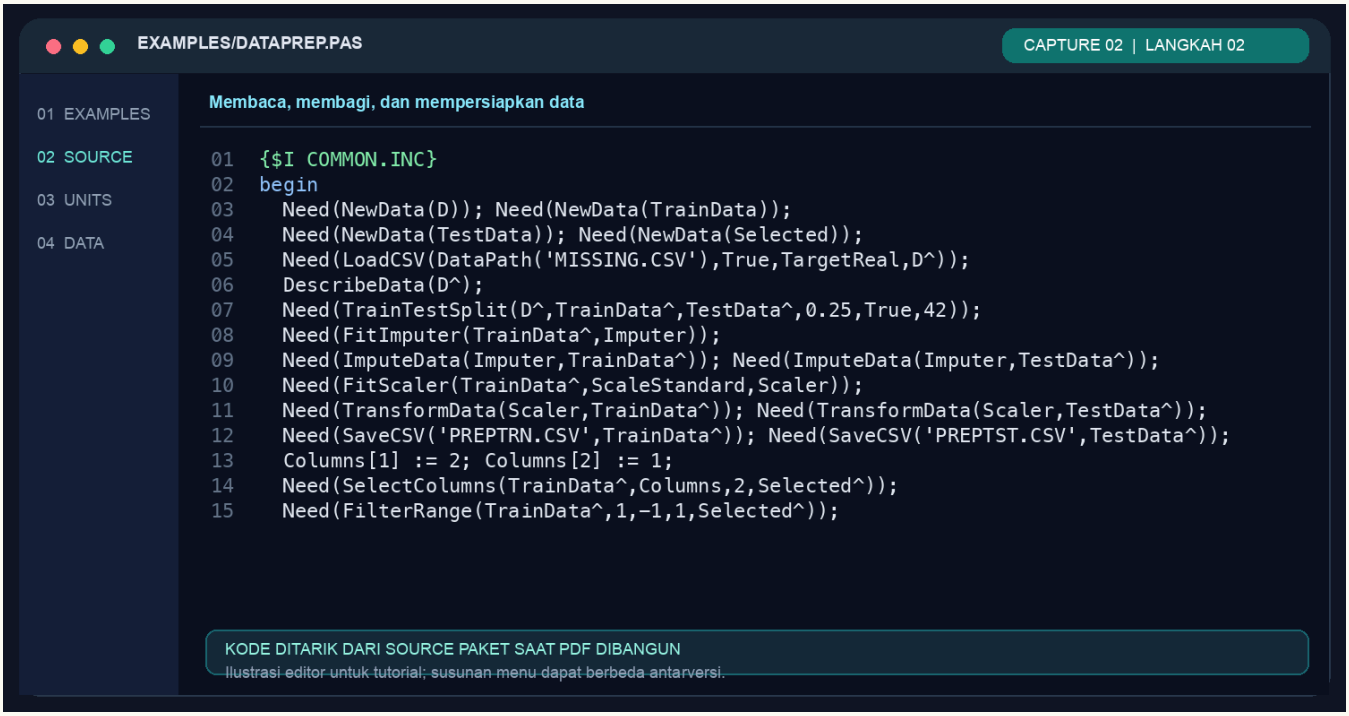
DATAPREP membaca MISSING.CSV, membagi train/test, mengimputasi, menstandarkan, lalu menulis dua CSV. FitImputer dan FitScaler menerima train; ImputeData dan TransformData menerapkan parameter yang sama ke train dan test.

Rujukan source: EXAMPLES/DATAPREP.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 02 — DATAPREP

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Periksa DATA/MISSING.CSV dan tandai kolom yang kosong.
2. Jalankan DATAPREP.PAS dari root otp-AI agar path DATA ditemukan.
3. Periksa jumlah baris train/test dan pesan kategori one-hot.
4. Buka PREPTRN.CSV serta PREPTST.CSV; pastikan keduanya memakai scaler train yang sama.



Gambar 02.1 — Visualisasi editor dengan cuplikan source DATAPREP.PAS.

Periksa hasil: FitImputer dan FitScaler menerima train; ImputeData dan TransformData menerapkan parameter yang sama ke train dan test.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Regresi linear satu fitur

Program REGDEMO.PAS | Unit: AIDATA + AISPLIT + AILINEAR + AISTATS + AIPLOT

Pelajari alur supervised learning: holdout split, fit model pada train, prediksi test, dan ukur galat tanpa menyamakan korelasi dengan sebab-akibat.

PERAN UNIT

AISPLIT membentuk holdout; AILINEAR mempelajari koefisien regresi; AISTATS mengukur galat pada data uji.

TUJUAN CONTOH

Menjalankan siklus supervised regression dari CSV sampai evaluasi, termasuk satu prediksi observasi baru dan grafik garis.

FUNGSI DAN PARAMETER UTAMA

TrainTestSplit(0.25, True, 42) berarti uji 25%, acak, seed tetap. FitLinear Ridge=0 berarti OLS tanpa penalti ridge; evaluasi memakai PredictLinearData(test); X baru=5.

HASIL YANG DIHARAPKAN

Mencetak jumlah train/test, intercept dan slope, MAE/RMSE, R2 jika terdefinisi, prediksi X=5, serta REGRESI.SVG.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Memisahkan fit dari evaluasi membantu mengukur generalisasi; data jam/nilai bersifat sintetis dan tidak menunjukkan sebab-akibat.

ALUR KODE DAN CATATAN

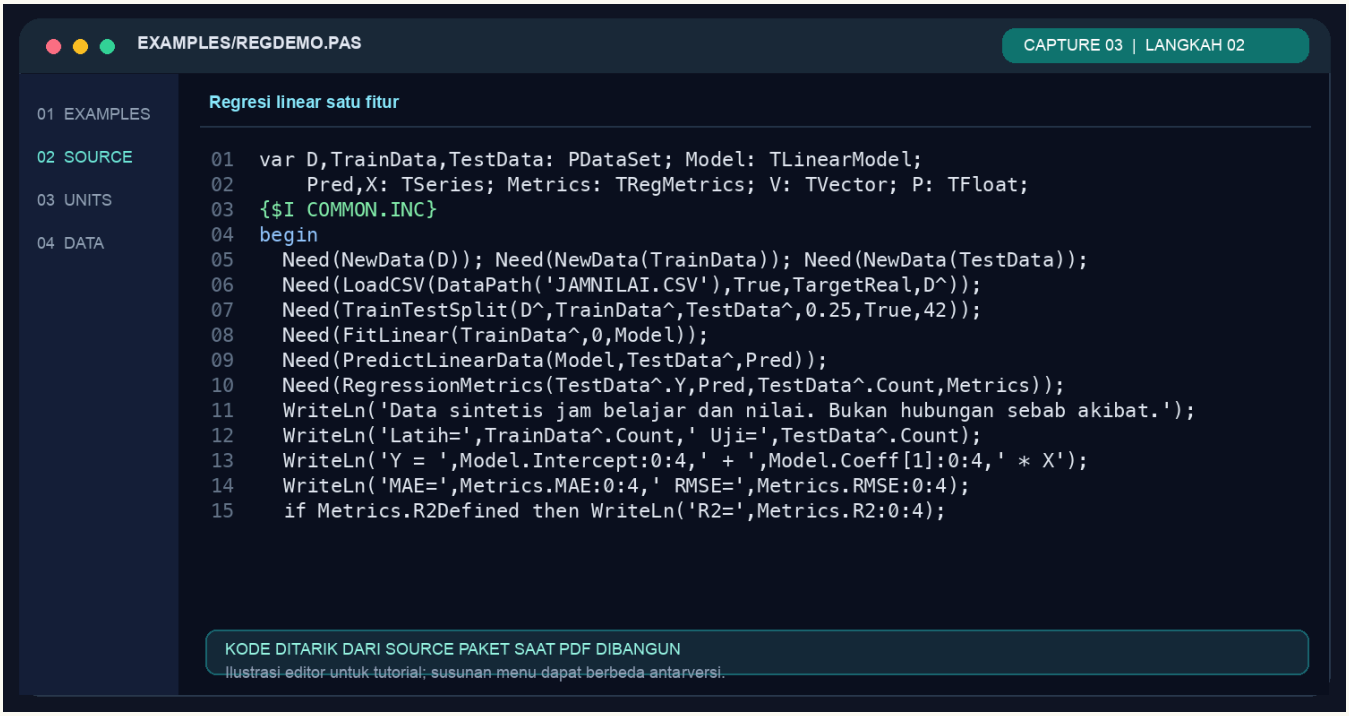
REGDEMO memakai data sintetis JAMNILAI.CSV dan mencetak persamaan, MAE/RMSE, serta prediksi satu observasi. MAE dan RMSE dinyatakan dalam satuan target; R2 hanya dicetak jika terdefinisi.

Rujukan source: EXAMPLES/REGDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 03 — REGDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Load JAMNILAI.CSV dan pastikan TargetReal digunakan.
2. Pisahkan 75% train dan 25% test dengan seed tetap.
3. FitLinear hanya pada train; evaluasi prediksi dengan target test.
4. Ubah seed lalu bandingkan hasil; jangan menyebut data sintetis sebagai bukti kausal.



Gambar 03.1 — Visualisasi editor dengan cuplikan source REGDEMO.PAS.

Periksa hasil: MAE dan RMSE dinyatakan dalam satuan target; R2 hanya dicetak jika terdefinisi.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Regresi multivariat dan koefisien

Program REGMULT.PAS | Unit: AIDATA + AILINEAR + AISTATS

REGMULT memulihkan persamaan buatan Y=5+2X1-3X2. Tujuannya memahami koefisien dan residual pada fixture yang dibuat deterministik.

PERAN UNIT

AILINEAR menyesuaikan satu model untuk satu atau banyak fitur; AISTATS menghitung metrik regresi dari target dan prediksi.

TUJUAN CONTOH

Membaca efek beberapa fitur sekaligus dengan membandingkan koefisien hasil model terhadap persamaan sintetis yang diketahui.

FUNGSI DAN PARAMETER UTAMA

FitLinear(D, Ridge=0): Ridge=0 meminta OLS; dua kolom X menghasilkan dua koefisien. PredictLinearData membuat prediksi tiap baris; RegressionMetrics menghitung RMSE latih.

HASIL YANG DIHARAPKAN

Intercept diharapkan mendekati 5, koefisien fitur mendekati 2 dan -3, serta RMSE mendekati nol karena fixture dibuat dari persamaan itu.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Mengenalkan representasi linear multivariat dan interpretasi bobot; kecocokan sempurna pada data pembangkit bukan bukti performa di luar sampel.

ALUR KODE DAN CATATAN

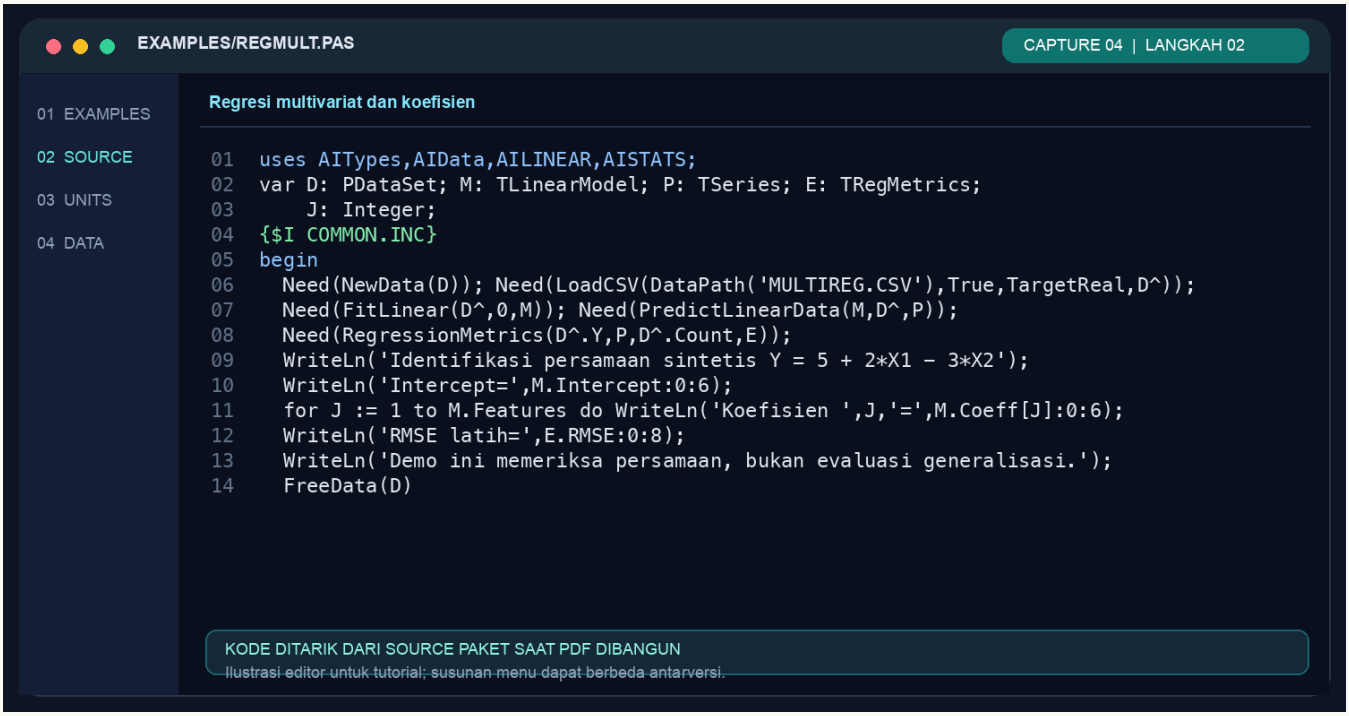
Model linear menerima beberapa fitur; koefisien dapat dibaca dari Model.Coeff[1..Features]. Koefisien mendekati 2 dan -3 serta intercept 5 untuk data sintetis ini.

Rujukan source: EXAMPLES/REGMULT.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 04 — REGMULT

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Buka MULTIREG.CSV untuk melihat dua fitur dan satu target.
2. Fit model dengan Ridge=0, lalu cetak intercept dan tiap koefisien.
3. Bandingkan koefisien dengan persamaan pembentuk fixture.
4. Ingat: RMSE latih pada data sintetis bukan ukuran generalisasi.



Gambar 04.1 — Visualisasi editor dengan cuplikan source REGMULT.PAS.

Periksa hasil: Koefisien mendekati 2 dan -3 serta intercept 5 untuk data sintetis ini.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

K-nearest neighbors untuk klasifikasi

Program KNNDEMO.PAS | Unit: AIDATA + AISPLIT + AIKNN + AISTATS + AIPLOT

KNN menyimpan pointer ke dataset train. Scaling perlu di-fit pada train; dataset tersebut harus tetap hidup dan tidak berubah selama model dipakai.

PERAN UNIT

AIKNN menyimpan dataset train dan mencari tetangga terdekat; AIDATA menskalakan fitur; AISTATS membuat confusion matrix dan skor.

TUJUAN CONTOH

Melakukan klasifikasi tiga kelas dengan KNN dan melihat bagaimana skala fitur, nilai K, serta tetangga memengaruhi prediksi.

FUNGSI DAN PARAMETER UTAMA

Split 0.30/seed 42 menyisihkan 30% uji; ScaleStandard dipelajari pada train; K=3 berarti tiga tetangga; Classes=3 mengatur ukuran confusion matrix.

HASIL YANG DIHARAPKAN

Mencetak confusion matrix dan accuracy/macro-F1 uji, kemudian menulis KNN.SVG dengan warna berdasarkan label prediksi.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Contoh supervised classification: K=3 menghaluskan keputusan lokal; scaling penting karena KNN mengandalkan jarak. BUNGA adalah data sintetis.

ALUR KODE DAN CATATAN

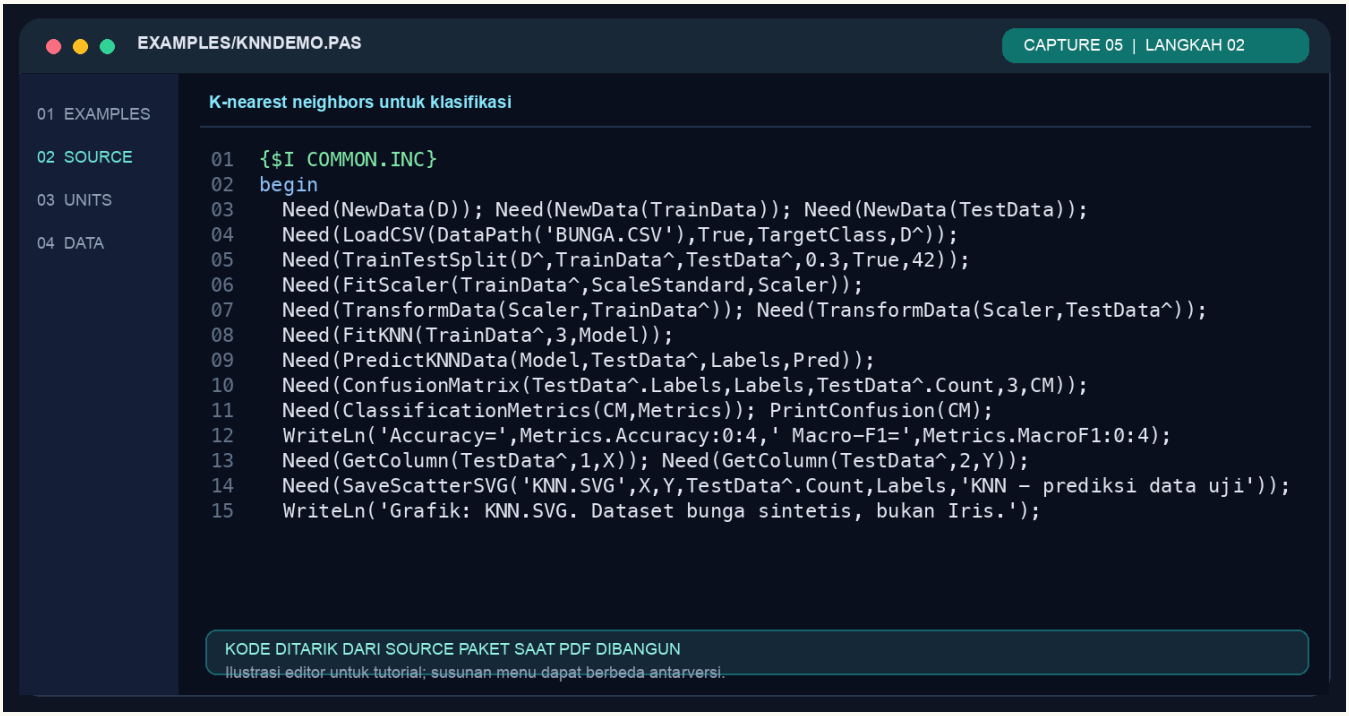
Demo menggunakan fixture bunga sintetis, K=3, confusion matrix, metrik kelas, dan ekspor scatter SVG. Dataset contoh sintetis, bukan Iris. Lepaskan referensi model sebelum dataset train dibebaskan.

Rujukan source: EXAMPLES/KNNDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 05 — KNNDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Load BUNGA.CSV dengan TargetClass dan lakukan split.
2. Fit scaler pada train; transform train dan test dengan scaler yang sama.
3. FitKNN pada train, lalu prediksi test dan hitung metrik.
4. Buka KNN.SVG hasil run; contoh keluaran terdahulu tersedia di OUTPUT/KNN.SVG.



Gambar 05.1 — Visualisasi editor dengan cuplikan source KNNDEMO.PAS.

Periksa hasil: Dataset contoh sintetis, bukan Iris. Lepaskan referensi model sebelum dataset train dibebaskan.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Clustering tanpa label

Program KMEANS.PAS | Unit: AIDATA + AIKMEANS + AIPLOT

K-means membentuk kelompok berdasarkan jarak, bukan label target. Nomor cluster tidak mempunyai urutan mutu dan bukan kelas aktual.

PERAN UNIT

AIKMEANS mengelompokkan baris berdasarkan pusat dan jarak; AIDATA menyediakan CSV dan scaling untuk membuat fitur sebanding.

TUJUAN CONTOH

Menemukan tiga kelompok tanpa memakai target kelas, kemudian membaca pusat cluster, konvergensi, dan inertia.

FUNGSI DAN PARAMETER UTAMA

FitScaler(ScaleStandard); FitKMeans(K=3, MaxIter=100, Tolerance=0.0001, Seed=42): K jumlah cluster, MaxIter batas putaran, Tolerance batas pergeseran pusat, Seed pengacak awal.

HASIL YANG DIHARAPKAN

Mencetak status konvergensi, inertia, tiga pusat; menyimpan CLUSTER.SVG serta INERTIA.SVG.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Memperkenalkan unsupervised learning: tidak ada label benar yang diberikan. Nomor cluster arbitrer; inertia bukan accuracy dan sensitif terhadap K/skala.

ALUR KODE DAN CATATAN

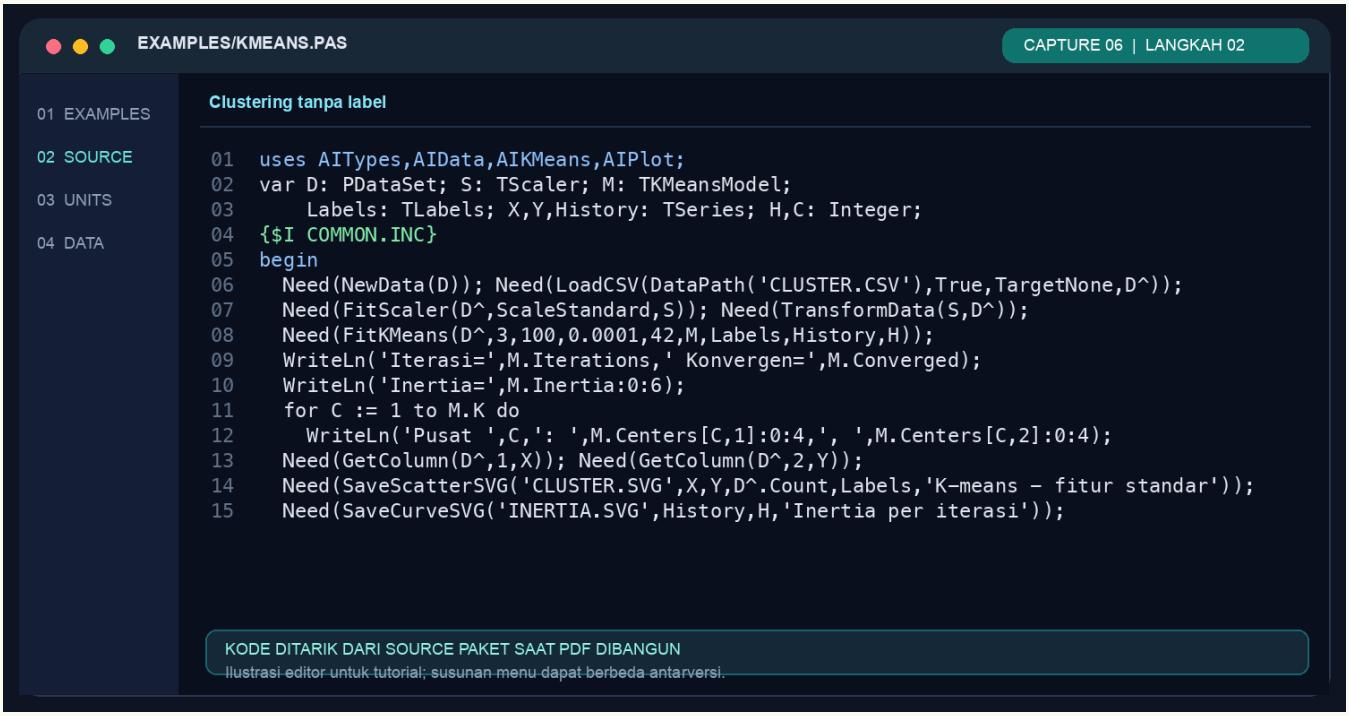
Contoh membandingkan pusat cluster dan inertia pada data geometris sintetis. Inertia adalah jumlah kuadrat jarak ke pusat terdekat; nilai lebih kecil tidak otomatis berarti model lebih baik.

Rujukan source: EXAMPLES/KMEANS.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 06 — KMEANS

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Buka DATA/CLUSTER.CSV dan catat bahwa cluster tidak disertai target kelas.
2. Pilih K yang masuk akal dan seed agar run dapat diulang.
3. FitKMeans, amati jumlah iterasi, status konvergensi, pusat dan inertia.
4. Bandingkan K=2, 3, dan 4; lihat OUTPUT/INERTIA.SVG.



Gambar 06.1 — Visualisasi editor dengan cuplikan source KMEANS.PAS.

Periksa hasil: Inertia adalah jumlah kuadrat jarak ke pusat terdekat; nilai lebih kecil tidak otomatis berarti model lebih baik.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Validasi silang K-fold

Program CROSSVAL.PAS | Unit: AIDATA + AISPLIT + AIKNN + AISTATS

K-fold menggunakan setiap baris sebagai data uji tepat satu kali. Pipeline preprocessing dan model perlu di-fit ulang secara terpisah pada setiap fold.

PERAN UNIT

AISPLIT menetapkan fold; AIKNN melatih/prediksi tiap fold; AIDATA melakukan scaling; AISTATS menghitung skor fold.

TUJUAN CONTOH

Membandingkan evaluasi holdout dengan lima evaluasi bergilir sambil memastikan scaler selalu di-fit ulang di dalam fold train.

FUNGSI DAN PARAMETER UTAMA

MakeFolds(N, Folds=5, Shuffle=True, Seed=42) menentukan lima bagian acak yang dapat diulang; FoldSplit menahan satu bagian; KNN memakai K=3 tetangga.

HASIL YANG DIHARAPKAN

Satu accuracy dicetak per fold lalu accuracy gabungan; program juga menyatakan bahwa fold acak ini tidak stratified.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Cross-validation memberi gambaran variasi akibat pemilihan sampel; preprocessing di dalam fold mencegah informasi test bocor ke train.

ALUR KODE DAN CATATAN

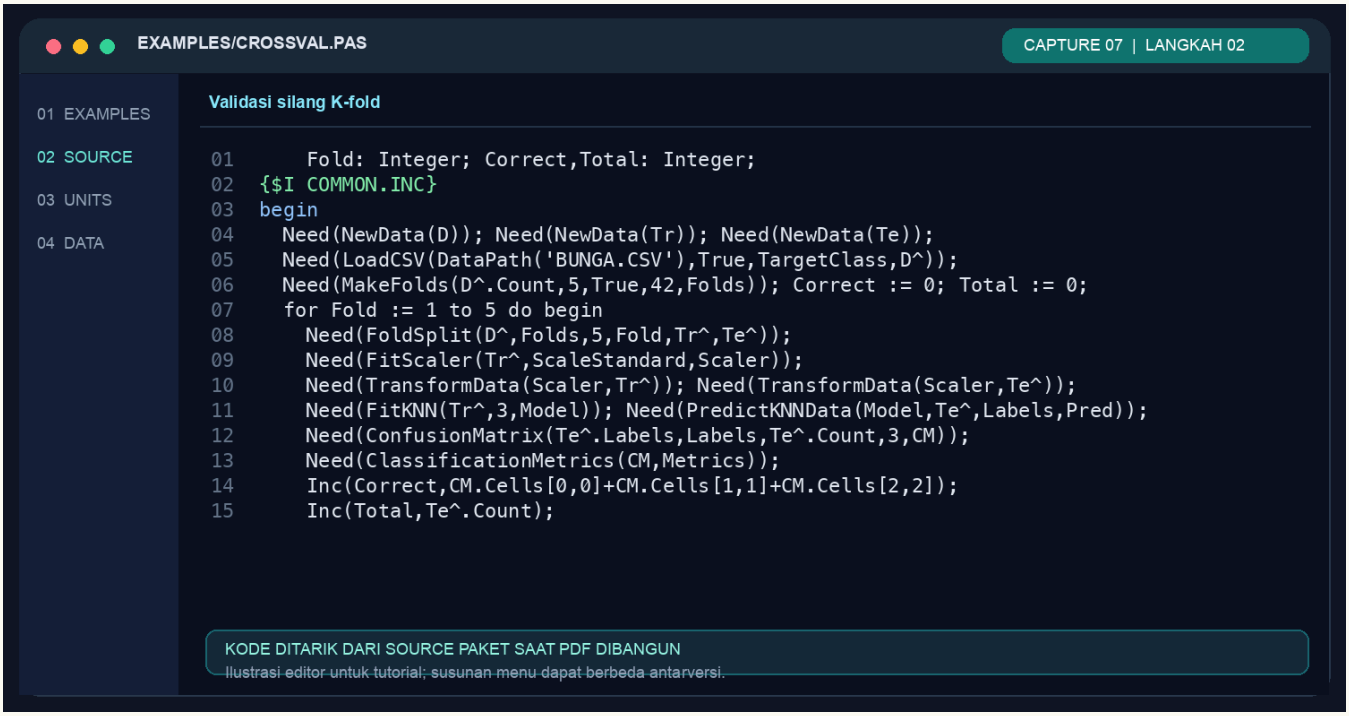
CROSSVAL membandingkan holdout dan beberapa fold dengan KNN pada fixture klasifikasi. AISPLIT tidak menyediakan stratifikasi kelas maupun time-series split.

Rujukan source: EXAMPLES/CROSSVAL.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 07 — CROSSVAL

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Atur jumlah fold dan seed; pilih Shuffle sesuai kebutuhan data.
2. Panggil MakeFolds lalu FoldSplit untuk setiap fold.
3. Fit transformasi/model hanya pada train fold, bukan keseluruhan dataset.
4. Ringkas metrik fold; jangan menganggap fold acak sesuai untuk data berurutan waktu.



Gambar 07.1 — Visualisasi editor dengan cuplikan source CROSSVAL.PAS.

Periksa hasil: AISPLIT tidak menyediakan stratifikasi kelas maupun time-series split.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Regresi logistik

Program LOGDEMO.PAS | Unit: AIDATA + AISPLIT + AILINEAR + AIOPTIM + AISTATS + AIPLOT

Model logistik menghasilkan probabilitas kelas biner. Fungsi loss memakai BCE pada logit; parameter optimisasi dan stopping harus diperiksa.

PERAN UNIT

AILINEAR berisi model logistik; AIOPTIM menyediakan konfigurasi training, clipping, momentum, dan stopping; AISTATS mengevaluasi klasifikasi.

TUJUAN CONTOH

Melatih classifier biner yang memetakan fitur ke probabilitas kelas 1 dan menilai label setelah menerapkan threshold.

FUNGSI DAN PARAMETER UTAMA

Split 0.25/seed 42; ScaleStandard train-only; DefaultTraining mengisi rate/epoch/momentum; FitLogistic melatih; threshold=0.5 membagi probabilitas menjadi kelas 0/1.

HASIL YANG DIHARAPKAN

Mencetak epoch, loss akhir, stop reason, accuracy, confusion matrix, dan menyimpan LOGLOSS.SVG.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Memperkenalkan sigmoid dan binary cross-entropy. Threshold mengubah trade-off keputusan, sementara metrik test mengukur hasil pada sampel terpisah.

ALUR KODE DAN CATATAN

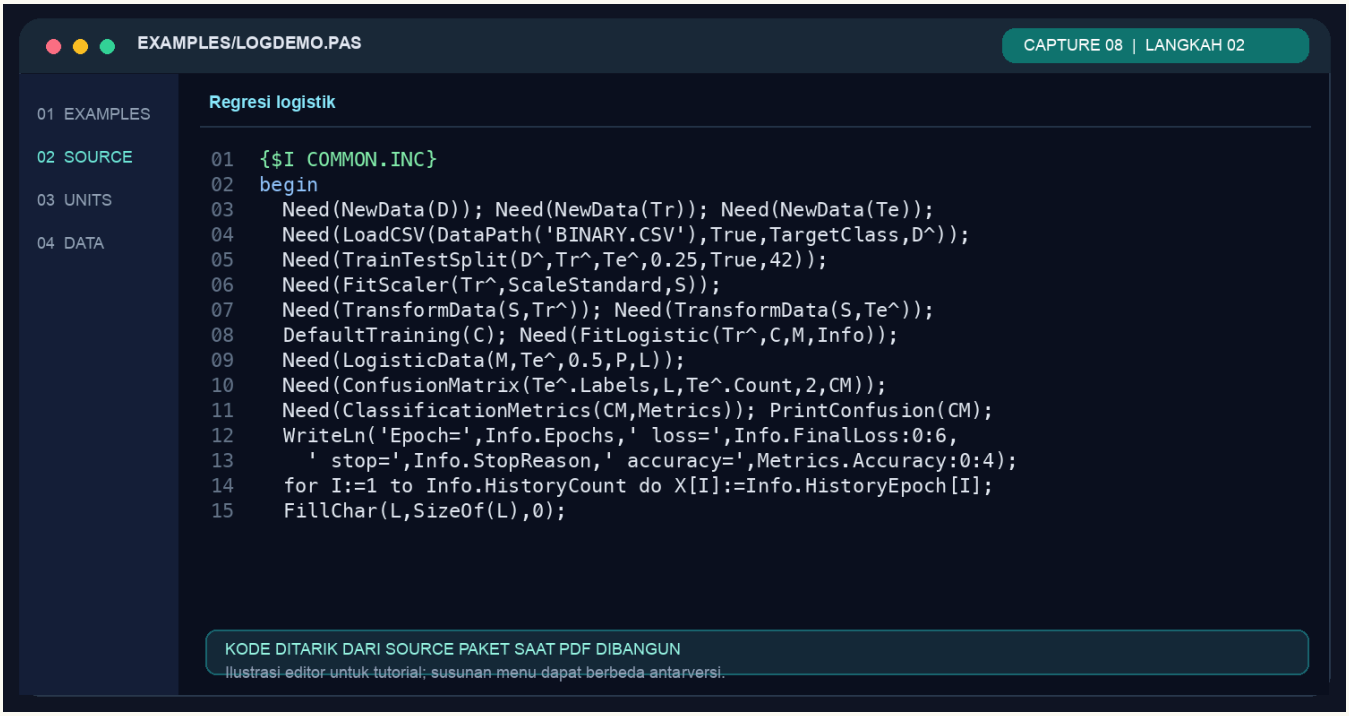
Demo melakukan split, scaling train, fit classifier, confusion matrix, dan membuat grafik loss. Seed tidak mengubah batch logistic yang deterministik. Converged hanya True pada stop goal.

Rujukan source: EXAMPLES/LOGDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 08 — LOGDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Load fixture kelas biner dan pastikan label hanya 0/1.
2. Fit scaler pada train; transform data train/test.
3. Atur konfigurasi melalui DefaultTraining, lalu fit model dan periksa Info.
4. Nilai probabilitas dan threshold menentukan kelas; amati LOGLOSS.SVG.



Gambar 08.1 — Visualisasi editor dengan cuplikan source LOGDEMO.PAS.

Periksa hasil: Seed tidak mengubah batch logistic yang deterministik. Converged hanya True pada stop goal.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Perceptron dan batas linear

Program PERDEMO.PAS | Unit: AIDATA + AILINEAR + AIOPTIM

Perceptron memperbarui bobot secara online dari kesalahan klasifikasi. AND dapat dipisahkan linear; XOR tidak bisa diselesaikan oleh satu perceptron.

PERAN UNIT

AILINEAR menyatukan model linear; AIOPTIM mendukung pola optimisasi; Perceptron mengklasifikasi dengan aturan pembaruan berbasis salah klasifikasi.

TUJUAN CONTOH

Membandingkan tugas AND yang dapat dipisahkan oleh garis dengan XOR yang tidak dapat dipisahkan oleh satu batas linear.

FUNGSI DAN PARAMETER UTAMA

FitPerceptron(MaxEpochs=100, Rate=0.1, Seed=42): batas epoch, besar langkah update, dan urutan acak. Dipanggil terpisah pada AND.CSV/XOR.CSV; score bukan probabilitas.

HASIL YANG DIHARAPKAN

AND diharapkan konvergen; XOR diharapkan tetap memiliki error dan tidak konvergen dengan perceptron tunggal.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Contoh menunjukkan kapasitas hipotesis: kegagalan linear pada XOR menjadi motivasi memakai fitur non-linear atau jaringan multi-layer.

ALUR KODE DAN CATATAN

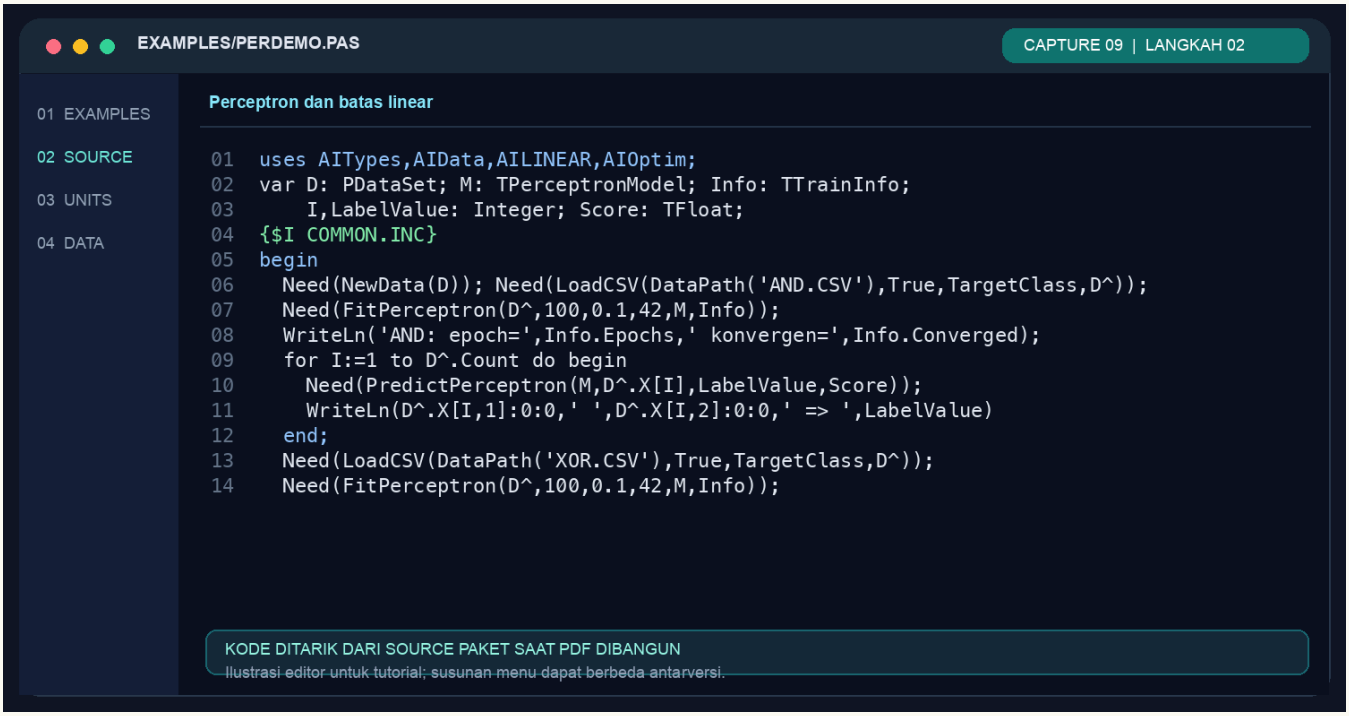
Contoh memperlihatkan konvergensi AND dan keterbatasan XOR sebagai konsep, bukan bug implementasi. Converged berarti tidak ada salah klasifikasi pada data latih saat evaluasi akhir epoch.

Rujukan source: EXAMPLES/PERDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 09 — PERDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Baca dataset AND dan XOR yang disediakan.
2. Atur seed dan konfigurasi training agar urutan contoh berulang.
3. FitPerceptron; periksa epochs, fraksi kesalahan, dan status konvergensi.
4. Bandingkan tabel prediksi AND dan XOR; jangan menafsirkan score sebagai probabilitas.



Gambar 09.1 — Visualisasi editor dengan cuplikan source PERDEMO.PAS.

Periksa hasil: Converged berarti tidak ada salah klasifikasi pada data latih saat evaluasi akhir epoch.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Jaringan saraf kecil untuk XOR

Program NNDEMO.PAS | Unit: AIDATA + AINNET + AIOPTIM + AIMODEL + AIPLOT

MLP OTP-AI adalah jaringan dense kecil dengan satu output biner. Contoh memudahkan melihat bobot, loss, stopping, prediksi, dan persistensi.

PERAN UNIT

AINNET menjalankan neural network kecil; AIOPTIM memvalidasi aturan SGD, momentum, clipping, loss, dan early stopping.

TUJUAN CONTOH

Belajar bagaimana hidden layer membuat XOR dapat dipelajari, membaca loss per epoch, lalu memeriksa penyimpanan model.

FUNGSI DAN PARAMETER UTAMA

FitNeural(Hidden=4); MaxEpochs=6000 adalah batas atas; Rate=0.5 ukuran langkah; Momentum=0.5 memakai velocity; Patience=1000; GoalLoss=0.01; seed default=42.

HASIL YANG DIHARAPKAN

Mencetak epoch/loss dan empat probabilitas/label prediksi; menulis XOR.TPM dan NNLOSS.SVG. Konvergensi tetap bergantung implementasi/seed.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Menunjukkan representasi bertingkat dan gradient-based learning. Model edukatif 2-4-1, bukan neural network produksi.

ALUR KODE DAN CATATAN

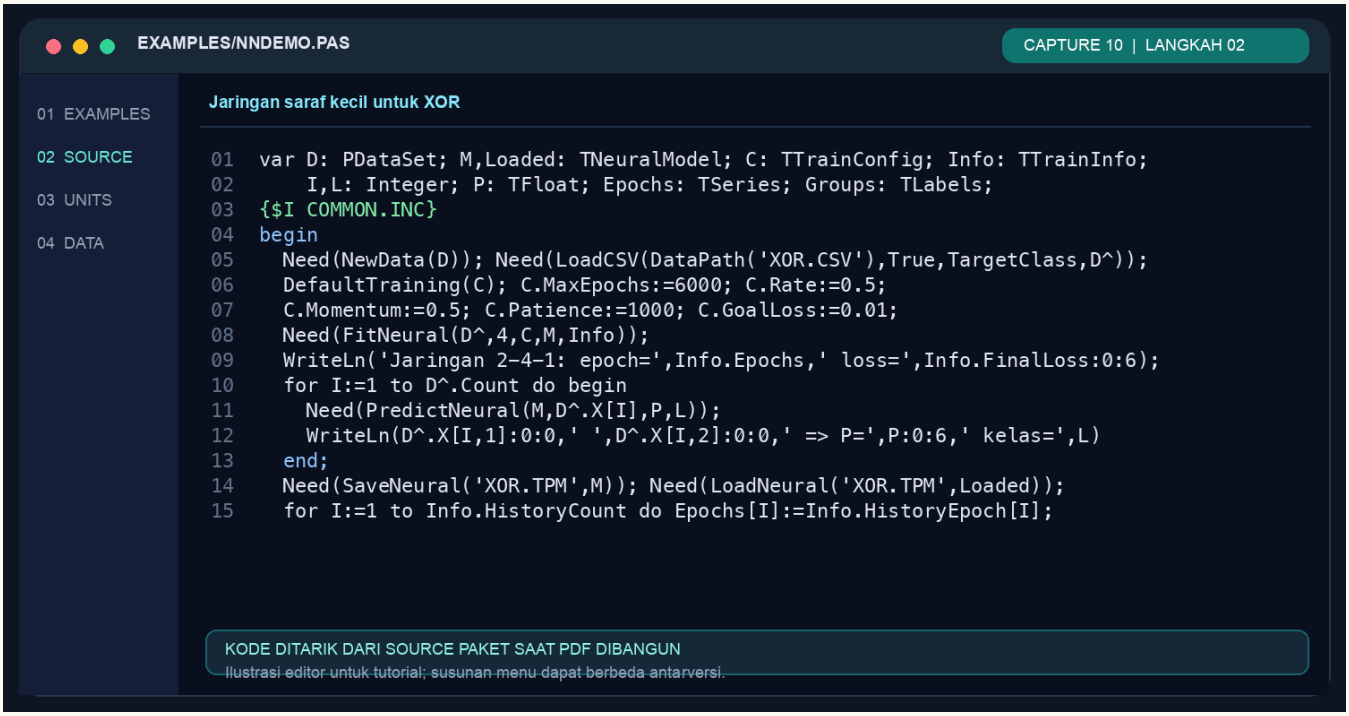
NNDEMO menggunakan XOR, empat neuron hidden, sigmoid, SGD online dan menyimpan XOR.TPM serta NNLOSS.SVG. Model edukatif terbatas: tidak ada autodiff, mini-batch, dropout, CNN, RNN atau transformer.

Rujukan source: EXAMPLES/NNDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 10 — NNDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Buka XOR.CSV dan verifikasi empat kombinasi input.
2. Ikuti konfigurasi Rate, Momentum, GoalLoss, Patience, MaxEpochs dan Seed.
3. FitNeural; cek prediksi per baris dan Info.FinalLoss.
4. Load model dari file hanya dengan compiler/ABI yang sama; lihat grafik loss.



Gambar 10.1 — Visualisasi editor dengan cuplikan source NNDEMO.PAS.

Periksa hasil: Model edukatif terbatas: tidak ada autodiff, mini-batch, dropout, CNN, RNN atau transformer.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Naive Bayes Gaussian

Program BAYDEMO.PAS | Unit: AIDATA + AISPLIT + AIBAYES + AISTATS

Gaussian Naive Bayes memodelkan mean dan varians fitur per kelas. VarianceFloor mencegah logaritma varians nol.

PERAN UNIT

AIBAYES menghitung Gaussian Naive Bayes dan posterior kelas; AISPLIT membentuk holdout; AISTATS meringkas prediksi.

TUJUAN CONTOH

Melatih distribusi fitur per kelas pada train dan mengubahnya menjadi prediksi serta probabilitas untuk sampel test.

FUNGSI DAN PARAMETER UTAMA

Split 0.30/seed 42; FitGaussian(VarianceFloor=0.000001) membatasi varians minimum; PredictBayes menghasilkan kelas/posterior; Classes=3 membangun confusion matrix.

HASIL YANG DIHARAPKAN

Mencetak confusion matrix dan Gaussian NB accuracy untuk test; probabilitas kelas disediakan ke setiap prediksi.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Memperkenalkan probabilitas kondisional dan asumsi Naive Bayes bahwa fitur independen bersyarat pada kelas.

ALUR KODE DAN CATATAN

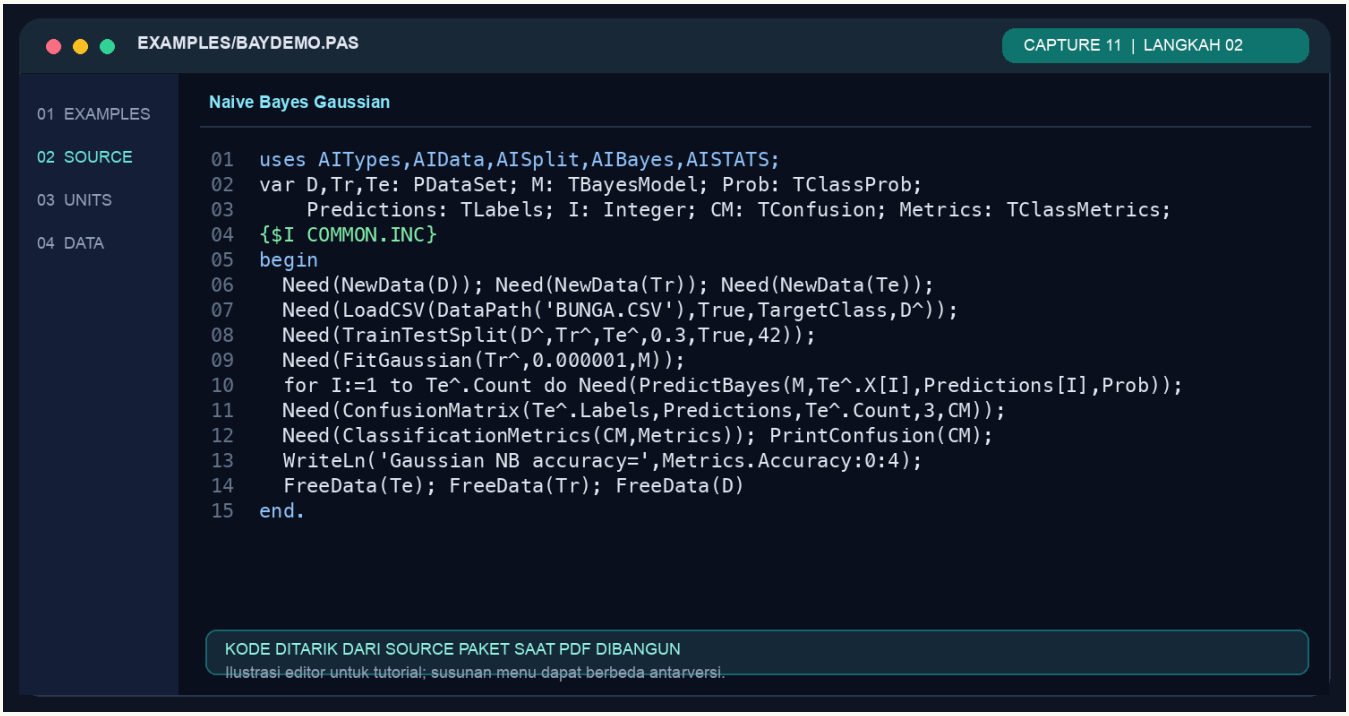
BAYDEMO menggunakan klasifikasi numerik sintetis dan mengamati posterior tiap kelas. Probabilitas model bukan jaminan kalibrasi; asumsi independensi fitur adalah penyederhanaan.

Rujukan source: EXAMPLES/BAYDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 11 — BAYDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Load data berlabel TargetClass.
2. Pilih VarianceFloor yang valid dan fit model.
3. Prediksi sampel; baca label dan Probabilities untuk label 0..9.
4. Periksa bahwa posterior berjumlah kira-kira satu dan kelas tak teramati bernilai nol.



Gambar 11.1 — Visualisasi editor dengan cuplikan source BAYDEMO.PAS.

Periksa hasil: Probabilitas model bukan jaminan kalibrasi; asumsi independensi fitur adalah penyederhanaan.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Teks, token, dan TF-IDF

Program TEXTDEMO.PAS | Unit: AITEXT + AIBAYES + AISTATS + AIMODEL

Pipeline teks ASCII kecil mengubah TSV menjadi token, vocabulary terpilih, vektor, lalu klasifikasi multinomial. Fit vocabulary hanya pada corpus train.

PERAN UNIT

AITEXT mengubah teks menjadi token dan fitur bag-of-words/TF-IDF; AIBAYES melatih classifier multinomial; AISTATS menghitung hasil uji.

TUJUAN CONTOH

Melatih vocabulary hanya dari corpus train, memvektorisasi teks, lalu mengklasifikasi pesan test tanpa memasukkan kata baru ke vocabulary.

FUNGSI DAN PARAMETER UTAMA

FitText Limit=8 membatasi fitur, MinDocumentFreq=1 menerima kata yang muncul minimal sekali, UseIDF=True memberi bobot IDF, Normalize=True memakai norm L2; Alpha=1 untuk smoothing.

HASIL YANG DIHARAPKAN

Mencetak kata vocabulary dan IDF, prediksi tiap teks beserta probabilitas, accuracy test, lalu menyimpan VOCAB.TPM dan TEXTNB.TPM.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Memperkenalkan representasi teks menjadi vektor numerik dan mencegah leakage vocabulary. Cakupan terbatas pada teks ASCII dan vocabulary kecil.

ALUR KODE DAN CATATAN

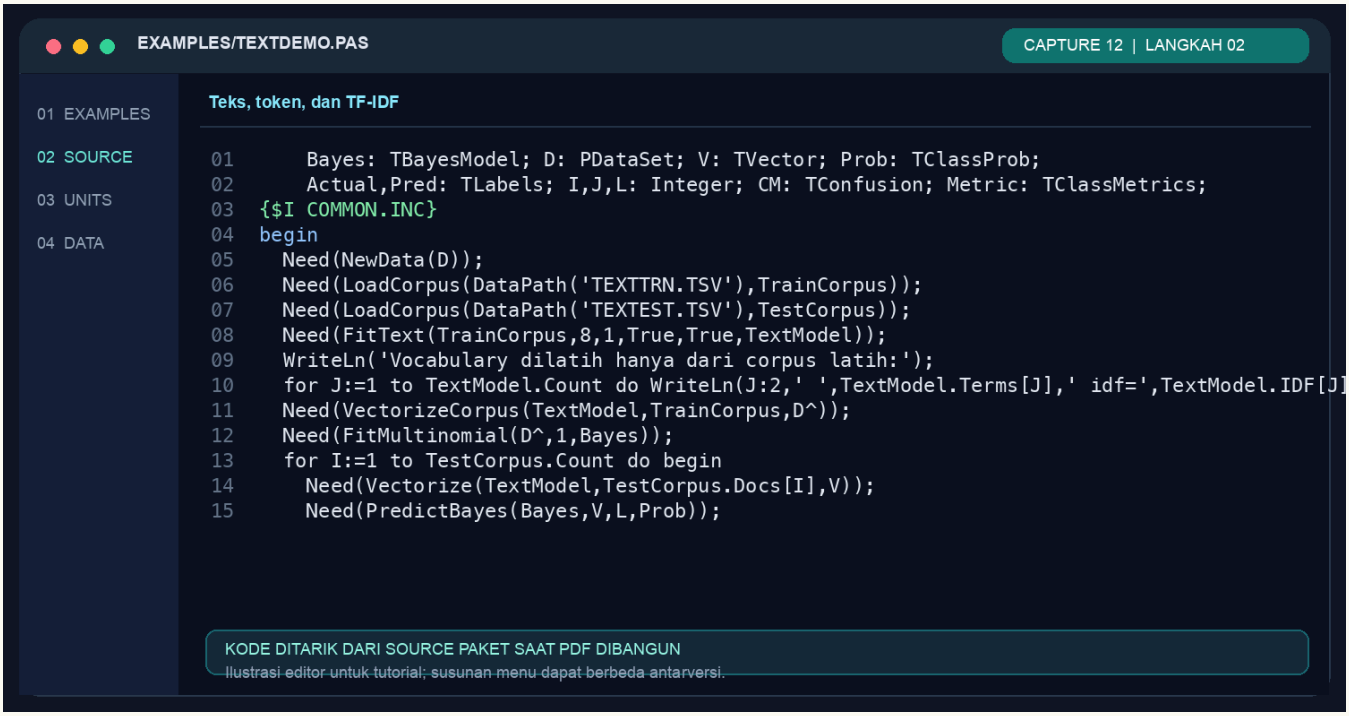
TEXTDEMO memakai TEXTTRN.TSV/TEXTEST.TSV dan Naive Bayes multinomial. Tidak ada Unicode, stemming, stopword, n-gram atau embedding; fitur dan vocabulary sengaja kecil.

Rujukan source: EXAMPLES/TEXTDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 12 — TEXTDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Periksa format TSV: label digit, TAB, kemudian teks ASCII.
2. Load corpus dan fit vocabulary dengan batas feature yang didukung.
3. Vectorize train/test memakai vocabulary train dan aturan IDF yang sama.
4. FitMultinomial lalu prediksi; amati kata baru yang diabaikan.



Gambar 12.1 — Visualisasi editor dengan cuplikan source TEXTDEMO.PAS.

Periksa hasil: Tidak ada Unicode, stemming, stopword, n-gram atau embedding; fitur dan vocabulary sengaja kecil.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Pohon keputusan

Program TREEDEMO.PAS | Unit: AIDATA + AITREE + AIMODEL

Pohon membagi fitur numerik memakai entropy dan information gain. Depth, MinLeaf, MinGain, serta batas 63 node membentuk kompleksitas.

PERAN UNIT

AITREE membangun pohon keputusan numerik; AIDATA membaca fixture; AIMODEL menyediakan simpan/muat snapshot pohon.

TUJUAN CONTOH

Menelusuri keputusan split dari root ke leaf serta mengamati bagaimana gain nol memungkinkan pohon memecah XOR.

FUNGSI DAN PARAMETER UTAMA

FitTree(DepthLimit=4 membatasi level, MinLeaf=1 jumlah minimum leaf, MinGain=0 mengizinkan split gain nol); TraceTree mengembalikan node yang dilalui dan label.

HASIL YANG DIHARAPKAN

Mencetak jumlah node, prediksi untuk empat kombinasi XOR, dan setiap aturan threshold yang dilalui menuju leaf.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Menghubungkan pemilihan split, entropy/information gain, kedalaman dan kompleksitas model dengan klasifikasi yang dapat diterangkan.

ALUR KODE DAN CATATAN

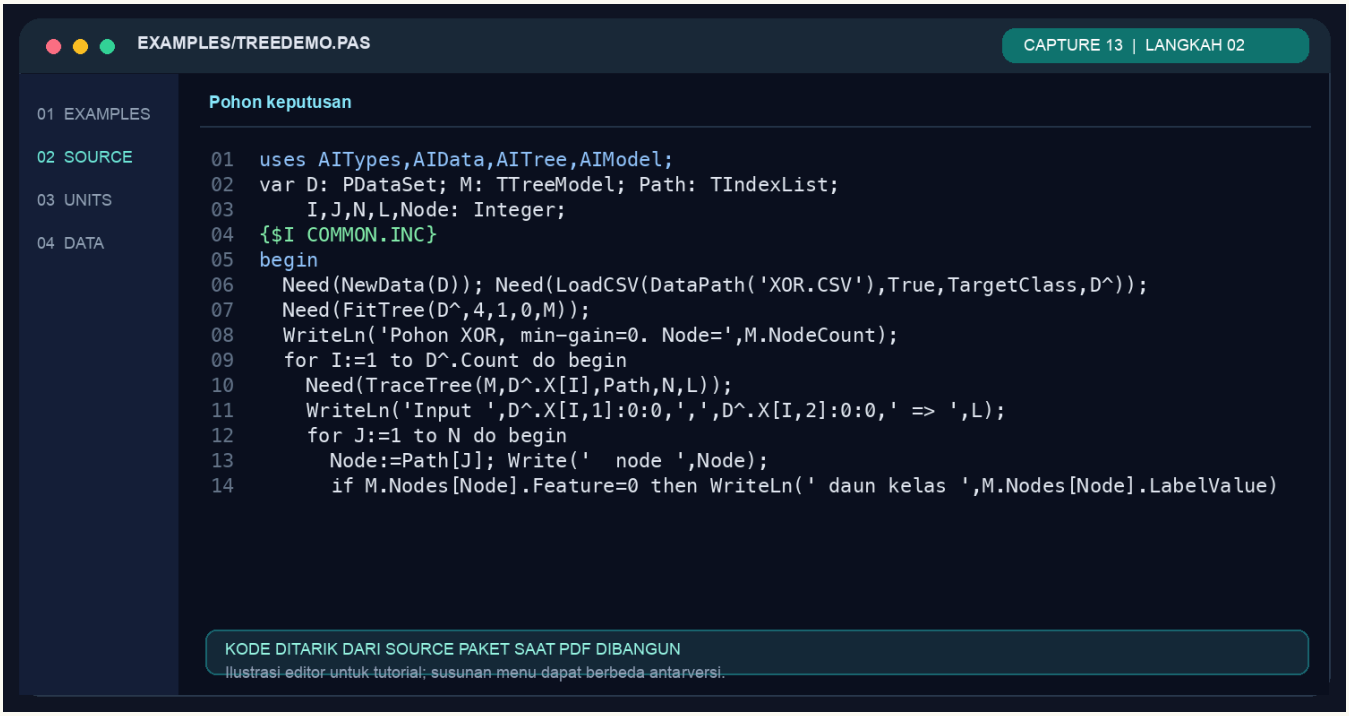
TREEDEMO memvisualisasikan prediksi/trace pada fixture kelas dengan batas kapasitas yang jelas. Tidak ada pruning, categorical split langsung, regression tree atau random forest.

Rujukan source: EXAMPLES/TREEDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 13 — TREEDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Load data numerik berlabel.
2. Pilih DepthLimit, MinLeaf, dan MinGain.
3. FitTree lalu periksa status Truncated dan ukuran pohon.
4. Gunakan TraceTree untuk melihat node dari akar hingga leaf.



Gambar 13.1 — Visualisasi editor dengan cuplikan source TREEDEMO.PAS.

Periksa hasil: Tidak ada pruning, categorical split langsung, regression tree atau random forest.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Sistem aturan dan forward chaining

Program RULEDEMO.PAS | Unit: AIRULE + AIMODEL

Aturan menurunkan fakta dalam putaran sampai tidak ada fakta baru. Provenance membantu menerangkan fakta hasil inferensi.

PERAN UNIT

AIRULE menyimpan fakta dan aturan serta menjalankan forward chaining; AIMODEL menyimpan rules/facts ke snapshot lokal.

TUJUAN CONTOH

Menunjukkan inferensi simbolik: fakta internet dan listrik menghasilkan kesiapan perangkat, lalu kesiapan perangkat menghasilkan layanan kelas daring.

FUNGSI DAN PARAMETER UTAMA

AddFact memakai nilai 1=true dan -1=unknown; AddRule mendefinisikan indeks premis, nilai yang diharapkan, kesimpulan; Infer mengembalikan trace dan jumlah aturan aktif.

HASIL YANG DIHARAPKAN

Infer menyimpulkan perangkat siap dan kelas daring siap; RULES.TPM serta FACTS.TPM ditulis.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Bukan model statistik: contoh membantu membedakan knowledge-based AI berbasis aturan dari supervised learning berbasis contoh berlabel.

ALUR KODE DAN CATATAN

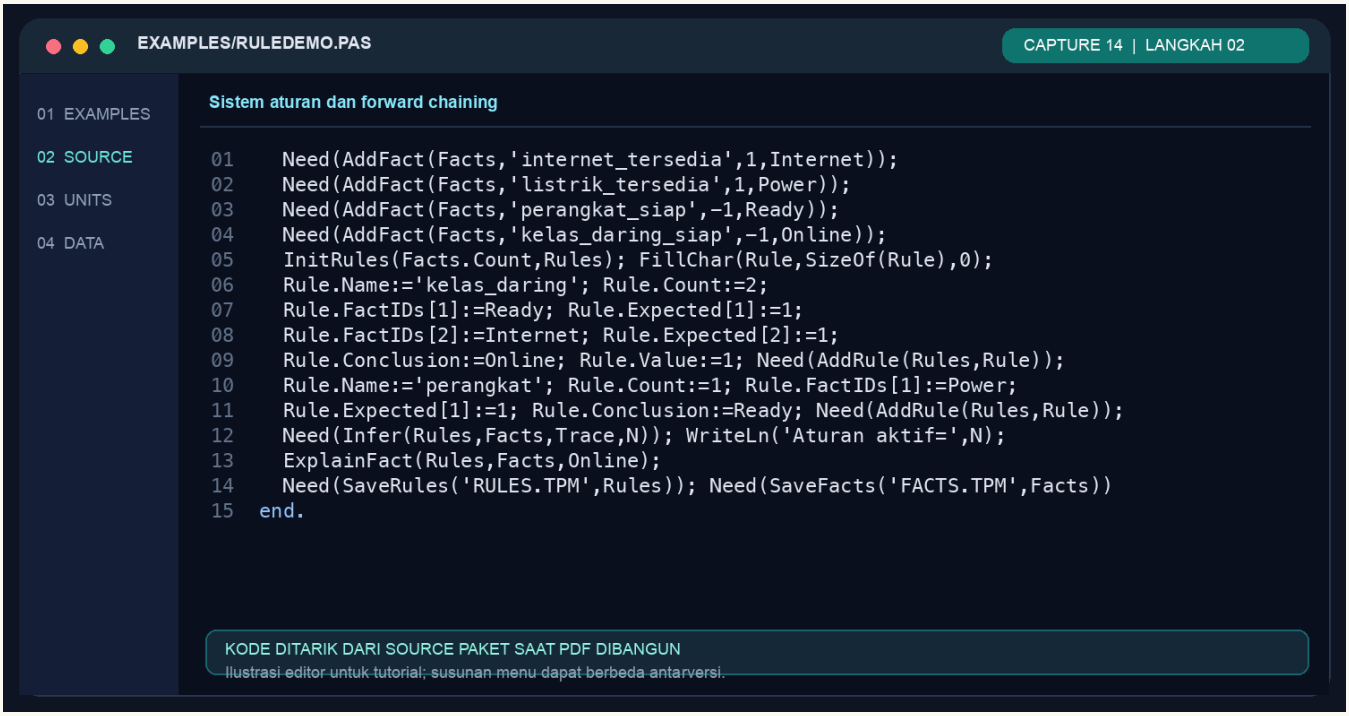
RULEDEMO mendefinisikan fakta dan aturan, menjalankan Infer, lalu menampilkan trace. Konflik aturan ditolak dan fakta tidak boleh ditimpa diam-diam.

Rujukan source: EXAMPLES/RULEDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 14 — RULEDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Inisialisasi fakta dengan nama dan nilai yang konsisten.
2. Tambahkan aturan setelah memeriksa premis dan kesimpulannya.
3. Jalankan Infer dan baca urutan firing.
4. Ubah fakta dasar; reset fakta turunan sebelum menjalankan ulang.



Gambar 14.1 — Visualisasi editor dengan cuplikan source RULEDEMO.PAS.

Periksa hasil: Konflik aturan ditolak dan fakta tidak boleh ditimpa diam-diam.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Pencarian jalur pada grid

Program PATHDEMO.PAS | Unit: AISEARCH

Grid kecil memperlihatkan BFS, DFS, dan A-star. Rute harus divalidasi terhadap dinding, batas grid, start dan goal.

PERAN UNIT

AISEARCH memodelkan grid, dinding, tetangga, dan algoritma BFS/DFS/A-star tanpa memakai dataset statistik.

TUJUAN CONTOH

Membandingkan strategi eksplorasi untuk menemukan jalur dari start ke goal dalam grid kecil dengan dinding.

FUNGSI DAN PARAMETER UTAMA

InitGrid(6,6); SetWall pada kolom 3 kecuali baris 4; SearchGrid(start=1, goal=36, method=BFS..AStar).

HASIL YANG DIHARAPKAN

Mencetak found, cost, jumlah node expanded dan gambar grid untuk BFS, DFS, dan A-star.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Mengajarkan pencarian state-space dan heuristic. Konsep ini berbeda dari belajar parameter model dari data.

ALUR KODE DAN CATATAN

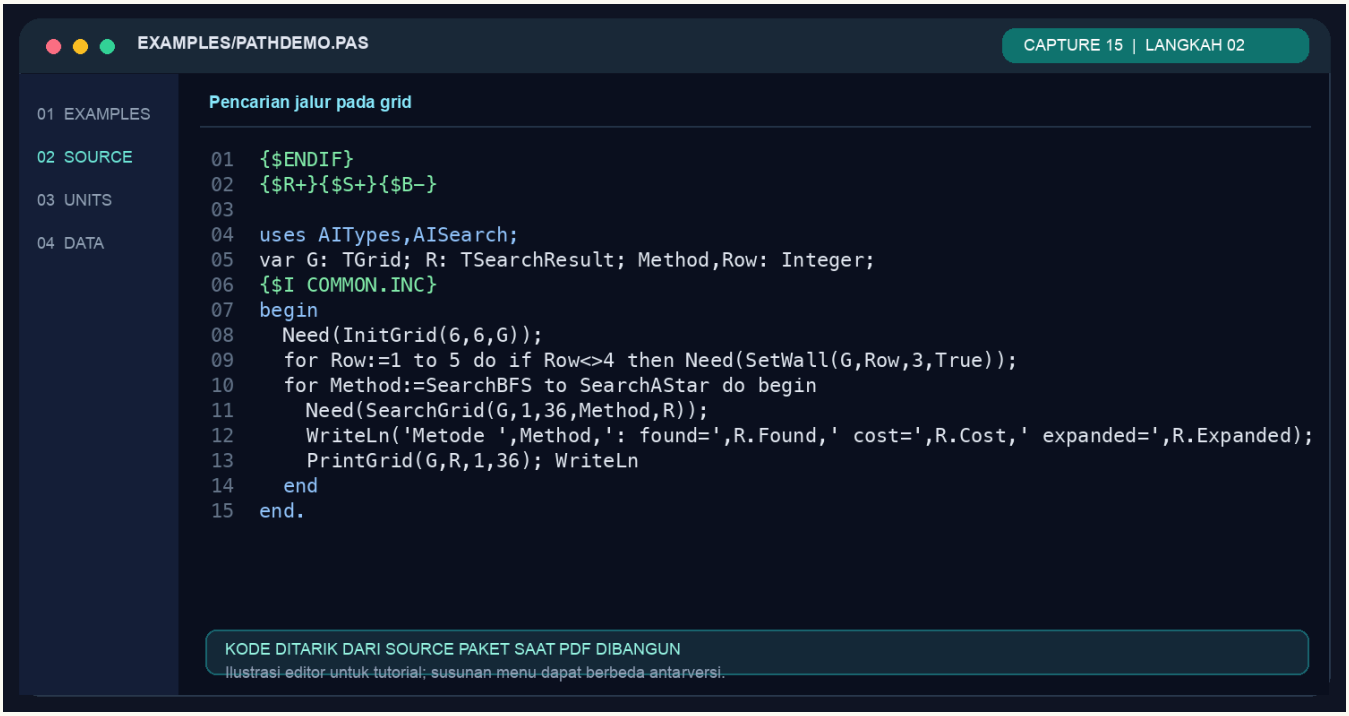
PATHDEMO membandingkan rute pada grid berhalang; A-star memakai heuristik grid. BFS menghasilkan rute terpendek pada biaya langkah seragam; DFS tidak menjamin rute terpendek.

Rujukan source: EXAMPLES/PATHDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 15 — PATHDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Inisialisasi ukuran grid dan tambahkan dinding.
2. Pastikan koordinat start dan goal berada dalam grid yang dapat dilalui.
3. Jalankan BFS, DFS, atau A-star.
4. Bandingkan panjang rute dan status Found; uji grid yang terputus.



Gambar 15.1 — Visualisasi editor dengan cuplikan source PATHDEMO.PAS.

Periksa hasil: BFS menghasilkan rute terpendek pada biaya langkah seragam; DFS tidak menjamin rute terpendek.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Inferensi fuzzy Mamdani

Program FUZDEMO.PAS | Unit: AIFUZZY + AIPLOT

Sistem fuzzy memetakan derajat keanggotaan input ke aturan dan centroid output. Nilai input harus berada pada universe yang dikonfigurasi.

PERAN UNIT

AIFUZZY mengelola fungsi keanggotaan segitiga, aturan Mamdani, firing strength dan centroid; AIPLOT menulis kurva hasil ke SVG.

TUJUAN CONTOH

Menerjemahkan suhu/kelembapan linguistik ke keluaran kontinu kecepatan kipas, lalu memplot respons saat suhu berubah.

FUNGSI DAN PARAMETER UTAMA

DefaultFuzzy; EvaluateFuzzy suhu=28, kelembapan=70, resolusi centroid=200; sapuan suhu X=0..40 dengan kelembapan 70.

HASIL YANG DIHARAPKAN

Mencetak kecepatan kipas dan sembilan firing strength; menghasilkan FUZZY.SVG untuk kurva respons.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Menunjukkan sistem aturan berbasis derajat keanggotaan yang dapat dibandingkan dengan klasifikasi/regresi yang belajar dari data.

ALUR KODE DAN CATATAN

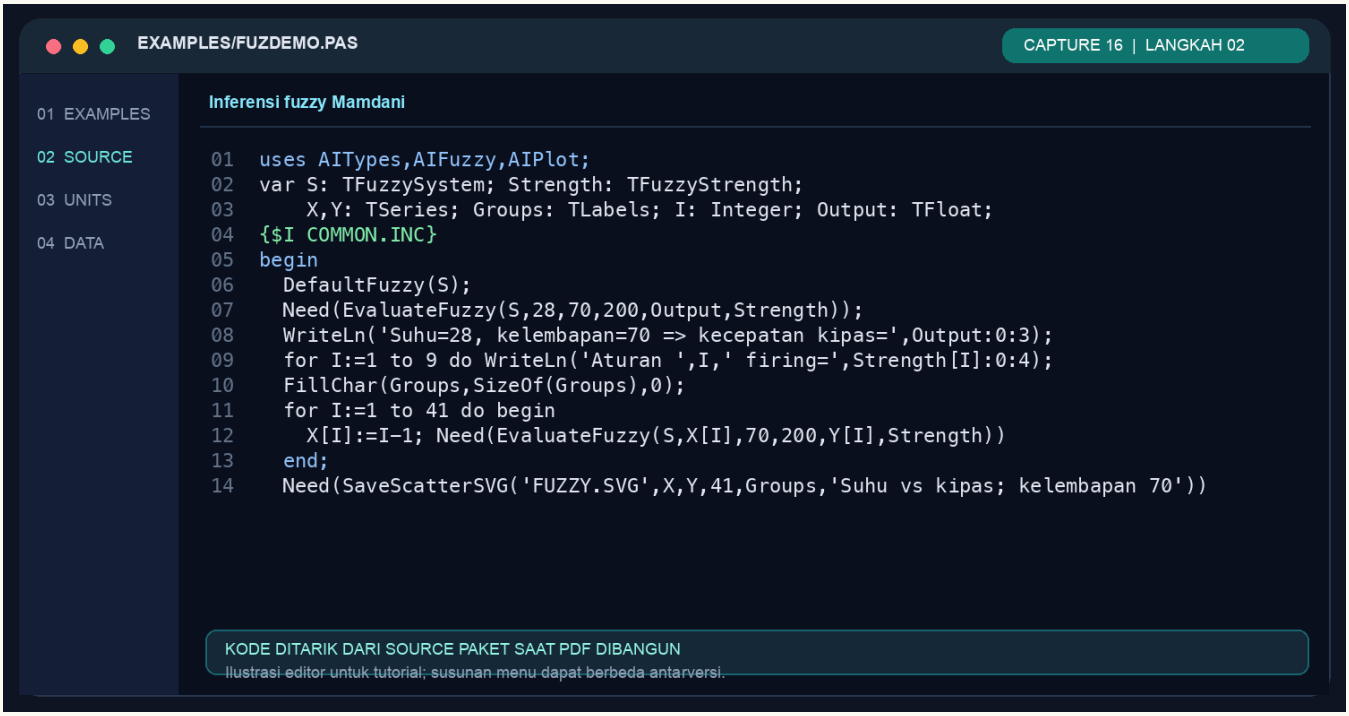
FUZDEMO memakai dua input, tiga himpunan dan aturan sederhana untuk menjelaskan defuzzifikasi. Sistem menolak segmen degenerate dan evaluasi tanpa rule aktif.

Rujukan source: EXAMPLES/FUZDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 16 — FUZDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Periksa rentang input/output dan parameter segitiga.
2. Susun aturan dengan indeks himpunan yang valid.
3. Evaluasi satu pasang input dan lihat kekuatan aturan.
4. Bandingkan kasus low, tengah, high, lalu uji input di luar universe.



Gambar 16.1 — Visualisasi editor dengan cuplikan source FUZDEMO.PAS.

Periksa hasil: Sistem menolak segmen degenerate dan evaluasi tanpa rule aktif.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Q-learning pada dunia grid

Program QDEMO.PAS | Unit: AIREWARD + AISEARCH + AIPLOT + AIMODEL

Q-learning memilih aksi dari state dan menerima reward/penalti. Terminal state tidak melakukan bootstrap dari nilai Q berikutnya.

PERAN UNIT

AIREWARD menyimpan Q-table dan memperbaruinya dari reward/transisi; AISEARCH menyediakan grid; AIPLOT merangkum return.

TUJUAN CONTOH

Melatih agent melewati grid berhalang, kemudian memeriksa policy greedy apakah mencapai goal dan membentuk rute.

FUNGSI DAN PARAMETER UTAMA

Grid 4×4; InitQ(16 state, 4 aksi, alpha=0.3 laju update, gamma=0.95 diskonto, epsilon=0.25 eksplorasi); TrainGridQ(start=1, goal=16, episodes=1500, maxSteps=100, seed=42).

HASIL YANG DIHARAPKAN

Mencetak jumlah episode sukses/langkah, found/cost rute greedy, grid, QRETURN.SVG, QTABLE.TPM, dan GRID.TPM.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Memperkenalkan reinforcement learning: agent belajar dari reward, eksplorasi epsilon, transisi, dan kondisi terminal—bukan label target per baris.

ALUR KODE DAN CATATAN

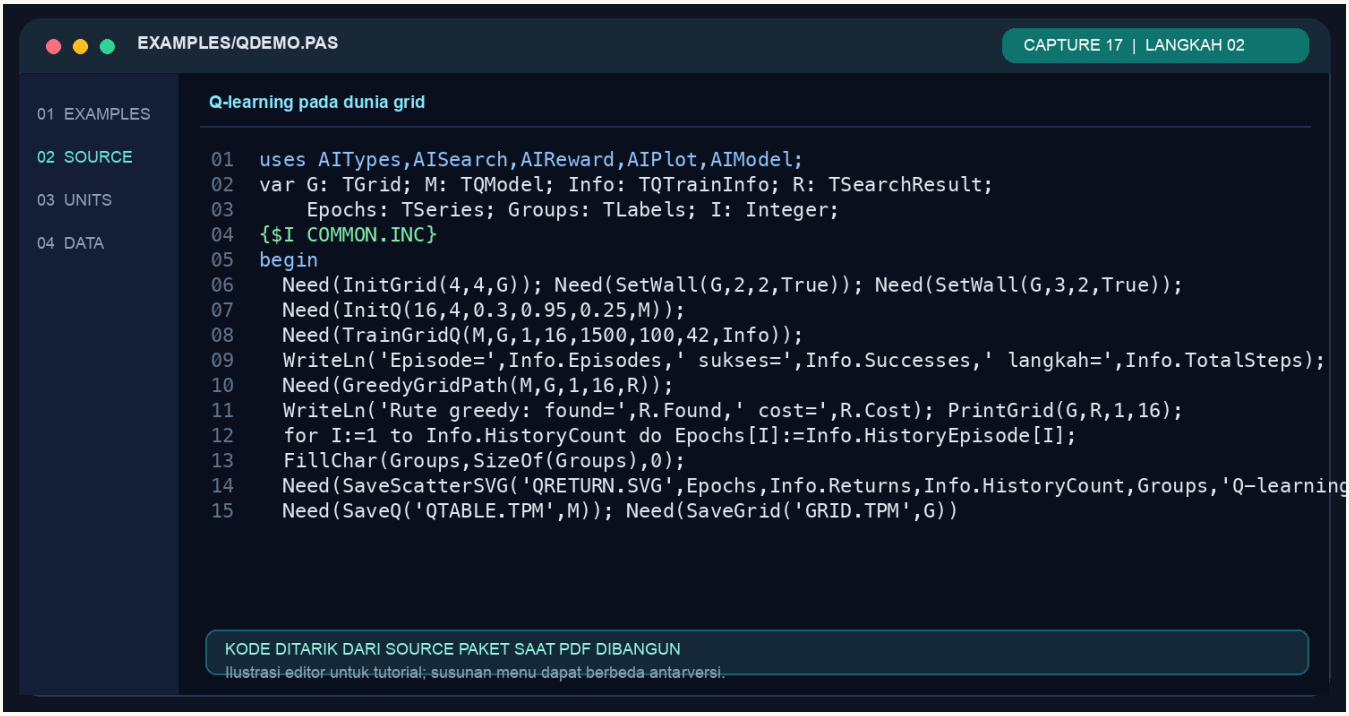
QDEMO melatih policy pada grid; hasilnya menekankan eksplorasi, epsilon decay, terminal reward, dan rute. Pelatihan dapat bergantung seed dan jumlah episode; periksa hasil, bukan hanya pesan training selesai.

Rujukan source: EXAMPLES/QDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 17 — QDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Bangun grid dengan start, goal dan dinding.
2. Atur alpha, gamma, epsilon, penalti, serta seed.
3. Jalankan training dan baca reward history serta policy greedy.
4. Verifikasi rute tidak terjebak siklus dan mencapai terminal.



Gambar 17.1 — Visualisasi editor dengan cuplikan source QDEMO.PAS.

Periksa hasil: Pelatihan dapat bergantung seed dan jumlah episode; periksa hasil, bukan hanya pesan training selesai.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Optimizer dan satu langkah SGD

Program OPTDEMO.PAS | Unit: AIOPTIM

Optimizer menyediakan konfigurasi, validasi, clipping, momentum, dan early stopping yang dipakai model lanjutan.

PERAN UNIT

AIOPTIM memvalidasi satu langkah SGD skalar/vektor dengan clipping dan momentum; model lain menggunakan prinsip serupa saat training.

TUJUAN CONTOH

Mengisolasi pengaruh gradient, velocity, learning rate, dan momentum pada fungsi sederhana sebelum melihat optimizer tersembunyi di model.

FUNGSI DAN PARAMETER UTAMA

Mulai w=-5, v=0; objective (w-3)^2; gradient 2(w-3); SGDStep(rate=0.1 learning rate, momentum=0.3 bobot velocity, clip=10 batas gradient); 100 langkah.

HASIL YANG DIHARAPKAN

Setiap 10 langkah nilai w dan loss dicetak; loss seharusnya turun menuju minimum w=3.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Membangun intuisi optimisasi numerik dan learning rate; parameter yang terlalu besar dapat mengganggu stabilitas update.

ALUR KODE DAN CATATAN

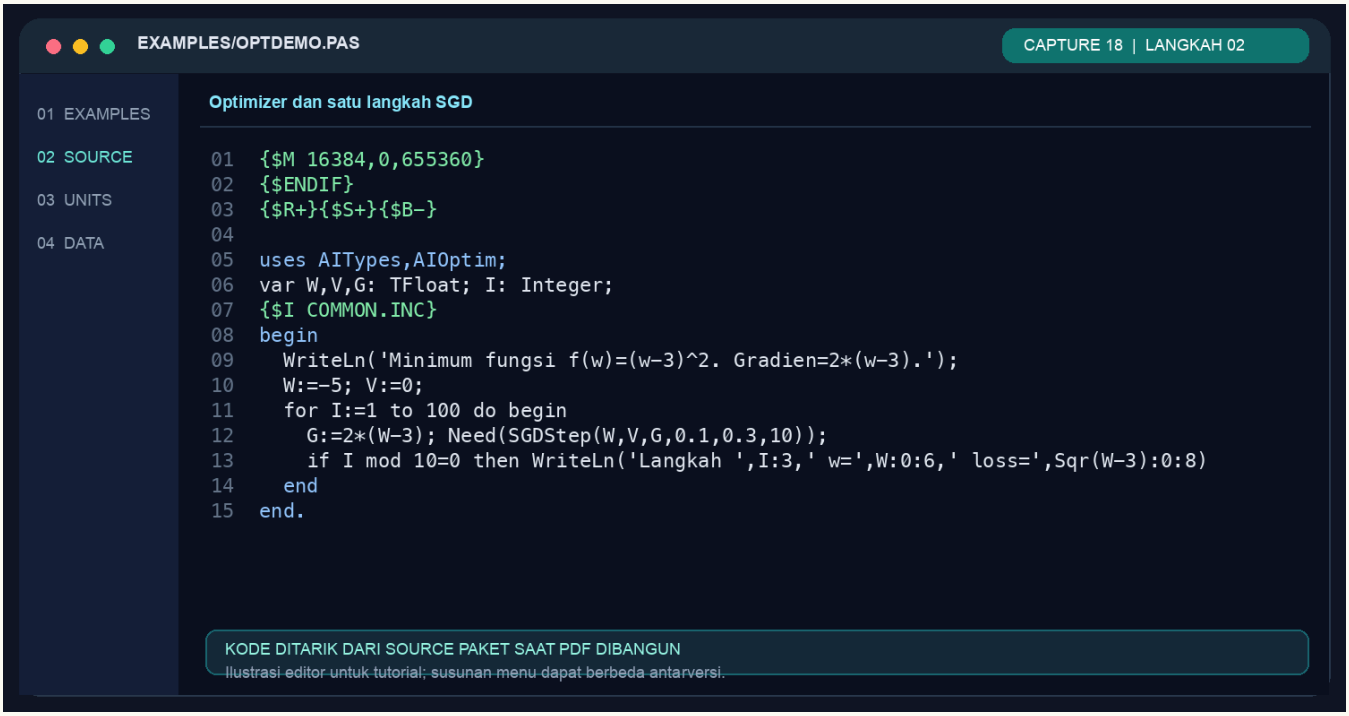
OPTDEMO mengisolasi update parameter agar pengaruh rate, gradient clipping, dan momentum terlihat. Clipping dilakukan per komponen dan L2 diterapkan oleh model, bukan SGDStep.

Rujukan source: EXAMPLES/OPTDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 18 — OPTDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Panggil DefaultTraining untuk mengisi nilai valid.
2. Catat value, velocity, gradient, rate dan momentum awal.
3. Jalankan SGDStep/VectorStep dan hitung pembaruan secara manual.
4. Uji konfigurasi invalid; pastikan fungsi gagal tanpa mutasi parsial.



Gambar 18.1 — Visualisasi editor dengan cuplikan source OPTDEMO.PAS.

Periksa hasil: Clipping dilakukan per komponen dan L2 diterapkan oleh model, bukan SGDStep.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Menyimpan dan memuat model

Program MODELIO.PAS | Unit: AIMODEL

AIMODEL menulis snapshot binary yang terikat ABI. Header, ukuran, endianness, layout dan checksum melindungi dari banyak berkas salah/rusak.

PERAN UNIT

AIMODEL memvalidasi dan menyimpan model; AILINEAR menyediakan regresi; AIKNN menunjukkan model yang terikat ke dataset storage.

TUJUAN CONTOH

Membandingkan prediksi sebelum/sesudah save-load dan memahami bahwa snapshot binary hanya aman untuk ABI/compiler yang sama.

FUNGSI DAN PARAMETER UTAMA

FitLinear(Ridge=0); prediksi X=5; FitKNN(K=3); LoadKNN membutuhkan Storage pemanggil untuk membind dataset model.

HASIL YANG DIHARAPKAN

Prediksi linear dan KNN sebelum/sesudah load diharapkan sama; file LINEAR.TPM dan KNN.TPM dibuat.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Model persistence menjaga parameter hasil training untuk pemakaian ulang; preprocessing, urutan fitur, dan ABI tetap bagian dari kontrak.

ALUR KODE DAN CATATAN

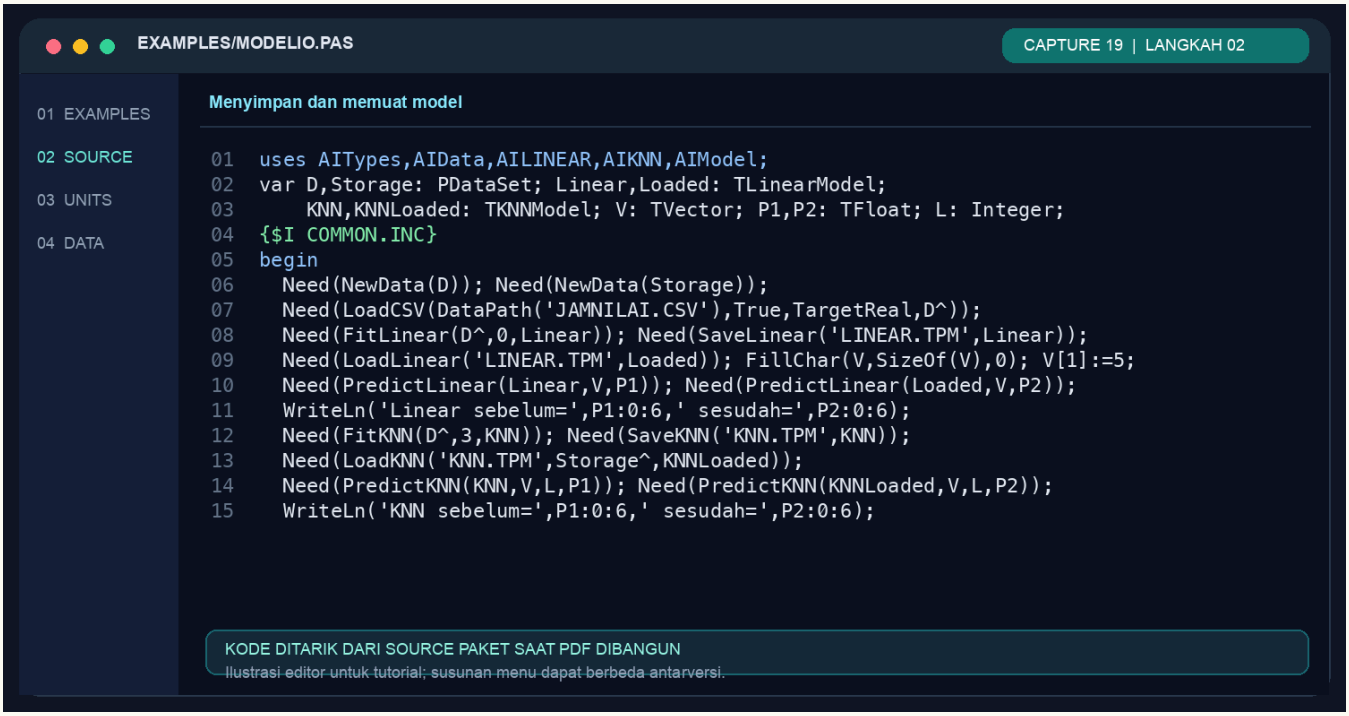
MODELIO menunjukkan pasangan Save/Load dan pentingnya storage pemanggil untuk KNN. Format .TPM bukan format interchange; header lama TPAIMDL2 dipertahankan sebagai identitas format.

Rujukan source: EXAMPLES/MODELIO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 19 — MODELIO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Fit model valid sebelum menyimpan.
2. Simpan ke path yang dapat ditulis; hindari mengandalkan file lama.
3. Load ke variabel baru, validasi hasil, lalu bandingkan prediksi.
4. Uji file rusak, truncated, trailing bytes, atau ABI berbeda.



Gambar 19.1 — Visualisasi editor dengan cuplikan source MODELIO.PAS.

Periksa hasil: Format .TPM bukan format interchange; header lama TPAIMDL2 dipertahankan sebagai identitas format.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

Grafik BGI dan SVG

Program PLOTDEMO.PAS | Unit: AIPLOT

AIPLOT mengekspor SVG tanpa dependensi tambahan. Backend layar BGI hanya tersedia untuk build Turbo Pascal dengan GRAPH dan driver yang sesuai.

PERAN UNIT

AIPLOT memvalidasi rentang data dan mengekspor scatter, kurva, serta garis regresi sebagai SVG; BGI hanya backend khusus compiler lama.

TUJUAN CONTOH

Menghubungkan nilai x-y dengan visualisasi yang dapat diperiksa tanpa Python, layanan web, atau pustaka plotting tambahan.

FUNGSI DAN PARAMETER UTAMA

20 titik; x=i/2; y=10+3x+(i mod 3-1); SaveRegressionSVG dengan intercept=10, slope=3, dan judul.

HASIL YANG DIHARAPKAN

PLOT.SVG dihasilkan; Odin menampilkan pesan bahwa BGI khusus Turbo Pascal dan SVG dibuka di peramban.

POSISI DALAM PEMBELAJARAN MACHINE LEARNING

Visualisasi membantu memeriksa pola, residual, dan cluster; plot bukan pengganti validasi kuantitatif pada data uji.

ALUR KODE DAN CATATAN

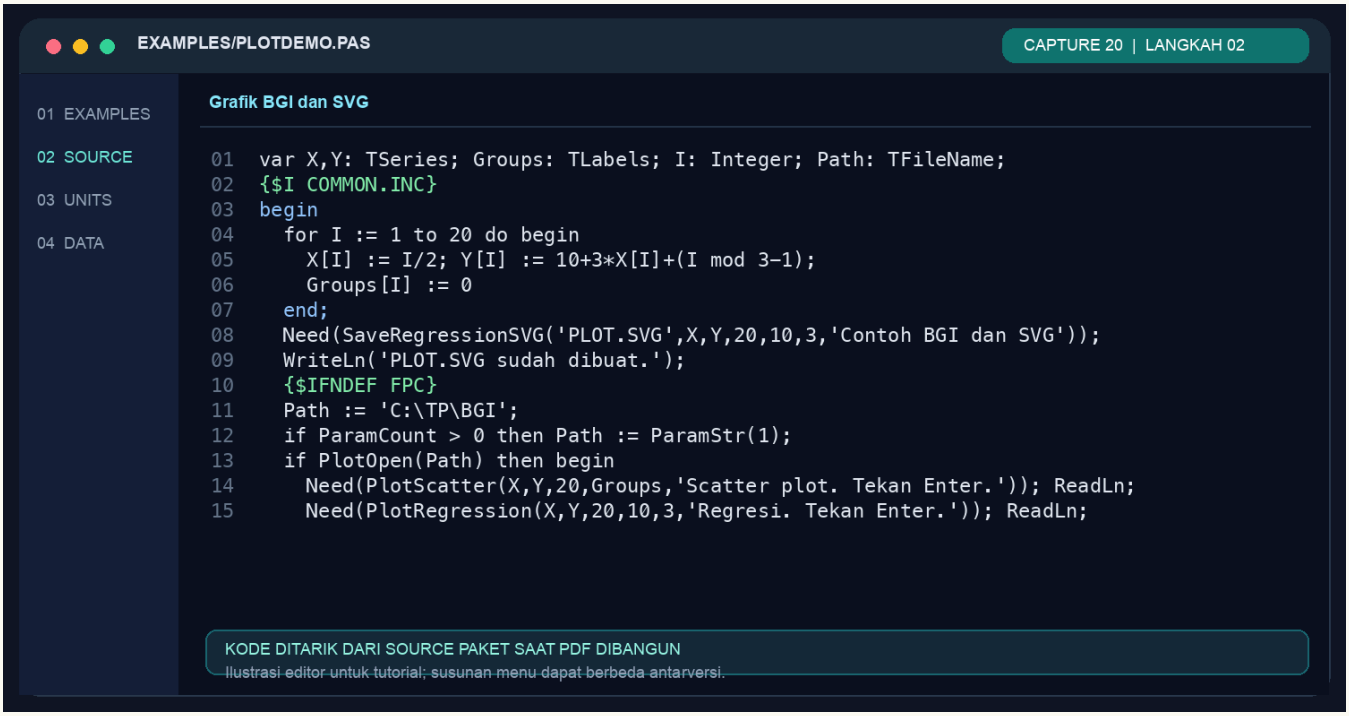
PLOTDEMO membedakan data, rentang sumbu, label, dan alur ekspor SVG dari layar BGI. Jika output SVG gagal ditulis, periksa AIError dan path; kegagalan tidak boleh dianggap sukses.

Rujukan source: EXAMPLES/PLOTDEMO.PAS. Cuplikan pada halaman berikut diambil dari file tersebut.

Praktik 20 — PLOTDEMO

Ikuti langkah satu per satu. Nomor di ilustrasi menunjukkan urutan interaksi; sesuaikan detail menu jika build IDE Anda berbeda.

1. Jalankan PLOTDEMO dengan working directory otp-AI.
2. Pada Odin/FPC, pilih fungsi Save*SVG dan buka hasil di peramban.
3. Periksa title, jumlah titik, label, dan rentang sumbu.
4. Gunakan PlotOpen hanya pada build Turbo Pascal yang menyediakan GRAPH/BGI.



Gambar 20.1 — Visualisasi editor dengan cuplikan source PLOTDEMO.PAS.

Periksa hasil: Jika output SVG gagal ditulis, periksa AIError dan path; kegagalan tidak boleh dianggap sukses.

Catatan: source pada gambar diambil langsung dari file saat PDF dibangun. Screenshot editor dibuat sebagai ilustrasi, bukan bukti tampilan atau menu tertentu.

LAMPIRAN A

Peta 19 unit canonical

Nama berikut cocok dengan file yang ada pada otp-AI/UNITS.

AITYPES	Tipe bersama, batas, dan error.
AIMATH	Numerik, jarak, random, dan matriks.
AISTATS	Statistik dan metrik evaluasi.
AIDATA	CSV, imputer, scaler, kategori, one-hot.
AISPLIT	Train/test split dan K-fold.
AILINEAR	Regresi linear/logistik dan perceptron.
AIKNN	K-nearest neighbors.
AIKMEANS	Clustering K-means.
AIOPTIM	SGD, momentum, clipping, stopping.
AINNET	Jaringan saraf kecil.
AIBAYES	Gaussian dan multinomial Naive Bayes.
AITREE	Pohon klasifikasi numerik.
AITEXT	Tokenisasi, corpus, dan TF-IDF.
AIRULE	Fakta, aturan, forward chaining.
AISEARCH	Grid search BFS/DFS/A-star.
AIFUZZY	Fuzzy Mamdani.
AIREWARD	Q-learning.
AIMODEL	Save/Load snapshot model.
AIPLOT	SVG dan backend grafik bersyarat.

> **Nama lama**
>
> Pada source sebelum konsolidasi, AIMATRIX→AIMATH, AIPREP→AIDATA, AIMETRIC→AISTATS, serta AIREG/AILOGREG/AIPERCEP→AILINEAR. Jangan menambahkan nama lama itu ke klausa uses contoh baru.

Diagnostik, batas, dan kebiasaan baik

Pesan error

Periksa nilai Boolean fungsi sebelum membaca output. Bila False, baca AIErrorCode dan AIError segera; panggilan berikut dapat mengganti status.

Dataset dan alokasi

Alokasikan TDataSet dengan NewData/FreeData. Dataset besar jangan diletakkan sebagai local stack variable. Hindari membebaskan storage KNN sebelum model tidak lagi digunakan.

Kapasitas

Dataset: 200 baris × 8 fitur; matriks: 16×16; model lain memiliki batas masing-masing. Batas bukan rekomendasi ukuran production.

Numerik

Input publik dibatasi 1E12. Operasi matriks add/subtract/scale/multiply menolak hasil di luar batas, mengembalikan ErrValue, dan tidak memutasi output.

SVG dan BGI

Pada Odin, ekspor SVG adalah jalur yang didukung. BGI bergantung pada Turbo Pascal, GRAPH.TPU, serta driver yang sesuai.

File .TPM

Snapshot binary terikat compiler, ABI, ukuran tipe, endianness, dan layout. Jangan memindahkannya lintas compiler.

Pengujian

Status terverifikasi di paket: Odin 2.5.0 berhasil compile 20 contoh, TESTALL 139 assertion, TESTADV 148 assertion. Turbo Pascal 7.0 asli dan BGI DOS belum diuji.

> **Bukan janji kinerja**
>
> Contoh dan fixture bersifat sintetis. Validasi keputusan model pada data dan lingkungan yang relevan; buku ini tidak mengklaim akurasi produksi.

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Version 3, 29 June 2007

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